

Everything You Need To Know About K-Theory

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Contents

1	Projective modules and Vector bundles	3
1.1	IBP, Stably Free, and Projective Modules	3
1.1.1	*Milnor Square	8
1.2	Picard Group of a Commutative Ring	11
1.3	Vector Bundles on Topological Spaces	16
2	K_0 Group	19
2.1	Algebraic Build-up	19
2.2	K_0 of a ring	20
2.3	Connection to H_0	25
2.3.1	Some Important Functionals	27
2.4	Morita Equivalence	31
2.5	*Mayer-Vietoris Sequence	33
2.6	Topological K-Group	34
2.6.1	K-groups over different fields	35
2.6.2	Bott Periodicity	36
3	K_1 Group	38
3.1	Defining K_1	38
3.2	K_1 of C^* -algebra	43
3.2.1	Basics of C^* algebras	43
3.2.2	K_1 theory of C^* algebras	46

All ring swill be associative with $1 \neq 0$. An R -module will be a right R -module unless specified. We do this so that a homomorphism $R^n \rightarrow R^m$ may be thought of as an $m \times n$ matrix with entries in R acting on column in R^n .

The field of K -theory emerged from the work of Alexander Grothendeick in trying to find invariants on Compact Hausdroff spaces and Schemes. The original idea tried to generalize natural transformations such as the determinant, and lead to a series of groups known as the K -groups which give important invariants that we shall explore. In the setting of Compact Hausdorff spaces, we even have some surprising results such as *Bott Periodicity* which gives a surprising result on the periodicity of the homotopy group of the Orthogonal group.

Projective modules and Vector bundles

If R is a field or a divisor ring, every R -module is a vector-space which is algebraically characterized up to dimension. More generally, a free R -module is a module that is R -module isomorphic to $R^{\oplus I}$ for some indexing set I . Any R -module homomorphism between two free modules is based on where chosen basis of the domain is represented by the chosen basis of the codomain; namely it can be represented by a matrix (or the generalization thereof for infinite dimension). These are the simplest modules that can be put over a ring. A natural question would be to see what are the possible “next weakest” conditions we can put, and what rings R “force” this condition. We shall explore a few of these conditions, one important one shall be for a module to be *projective* which we shall show how this could be thought of as being *locally free*. Measuring how this structure differs from being free will be the beginnings of the study of the K_0 -group in the next chapter.

1.1 IBP, Stably Free, and Projective Modules

If R is a field or a divisor ring, then $R^n \cong R^m$ implies $n = m$. If R is commutative this is still the case (consider $R/\mathfrak{m}$ by a maximal ideal), group rings (for finite groups), and any (left)-Noetherian ring satisfies this property as well (using a version of Nakayama’s lemma can prove this, see [2]). This property is *not* true for general rings:

Example 1.1: Failure of Invariant Basis Number

let R be a commutative ring and let M be the $\mathbb{N} \times \mathbb{N}$ matrices where each column has finitely many nonzero entries, this is sometimes denoted $CFM_{\mathbb{N}}(R)$. Then $CFM_{\mathbb{N}}(R) \cong CFM_{\mathbb{N}}(R)^2$ via the map:

$$\varphi : CFM_{\mathbb{N}}(R) \rightarrow CFM_{\mathbb{N}}(R)^2 \quad M \mapsto (\text{odd columns of } M, \text{ even columns of } M)$$

Another example the reader could work through is $\text{End}_k(k^{\oplus \mathbb{N}})$.

Definition 1.1.1: Invariant Basis Property

Let R be a ring. Then we say that R satisfies the (right) *invariant basis property* (IBP) if

$$R^n \cong R^m \quad \Rightarrow \quad n = m$$

If R satisfies IBP, then the rank of a free R -module M is an invariant, independent of the choice of basis.

Proposition 1.1.1: Necessary Condition For Ibp

Let R be a ring. Then R satisfies IBP if there exists an injective ring map $f : R \rightarrow F$ from R to a field (or division ring) F

Proof :

Given such a map, notice that any basis for a free R -module M is the basis for the vector-space $V = M \otimes_R F$, and V is independent of choice of basis, and hence any two bases must have the same cardinality.

Kernel's of a surjective R -module homomorphisms $\varphi : R^n \rightarrow R^m$ need not be free, however they must be stably free:

Definition 1.1.2: Stably Free

Let P be an R -module. Then P is called *stably free* of rank $n - m$ if there exists n, m such that

$$P \oplus R^m \cong R^n$$

If R satisfies IBP, then the rank of a stably free module is independent of the choice of n and m .

Certainly, the kernel stably free as R^m is free as we must have a split sequence (see [2]). Not all R -modules are stably free (or more interestingly, not all stably free modules are free). The notion of stably free seems very similar to *projective*, however we have not specified the condition for the 2nd summand in the definition of projective modules and hence it is a slightly weaker condition. We shall get to projective modules shortly.

Let us take a moment to consider the case where an R -module P is stably free of rank 1, that is $P \oplus R \cong R^n$. In this case, we may ask when can we “cancel” the R on each side, a fruitful discussion considering the K -theory we are building up to. In this case, let σ be a row vector of the right hand side, and call it the *unimodular row*. We get the following interesting set of equivalences:

1. σ is unimodular
2. $R^n \cong P \oplus R$ where $P = \ker(\pi)$ where $\pi : R^n \rightarrow R$ (given by σ)
3. $R = r_1 R + \cdots + r_n R$
4. $1 = r_1 s_1 + \cdots + r_n s_n$ for appropriate $s_i \in R$

Note that if P were free in $R^n \cong P \oplus R$, then a basis of P would give a new basis for R^n , which corresponds to an invertible matrix (say g) whose first row is the unimodular row $\sigma : R^n \rightarrow R$ corresponding to P . From this we get the following citation:

Proposition 1.1.2: Free and Unimodular

Let P be an R -module that stably free of rank 1. Then P is free if and only if the corresponding unimodular row may be completed to an invertible matrix.

Proof :

combine the above discussion.

If R is commutative, then every unimodular row of length 2 may be completed: if $r_1 s_1 + r_2 s_2 = 1$, then the invertible matrix is:

$$\begin{pmatrix} r_1 & r_2 \\ -s_2 & s_1 \end{pmatrix}$$

And so, if $R^2 \cong R \oplus P$, then $P \cong R$. We shall get stronger versions of this result later (note here though that commutativity was important, this fails for unimodular row of length to wit a non-commutative ring).

In dimension 3, we can already find counter example:

Example 1.2: Stably Free But Not Free

Take

$$R = \frac{\mathbb{R}[x, y, z]}{(x^2 + y^2 + z^2 - 1)}$$

Note that every element $(f, g, h) \in R^3$ gives a vector-field in $\mathbb{R}^3$. Let σ be the unimodular row $\sigma = (x, y, z)$. Then $1s$ is the vector-field pointing radially outward. If P were free, a basis of P would yield two tangent vector fields on S^2 which are linearly independent at every point of S^2 (since together with σ , they span the tangent space of 3-space at each point). But this contradicts the infamous Hairy-sphere theorem^a.

^aOr more humorously *Hairy ball theorem*

Let us now return to the discussion of projective modules.

Definition 1.1.3: Projective Module

Let P be an R -module. Then P is *projective* if there exists an R -module Q such that $P \oplus Q$ is a free R -module.

See [2] for details on the equivalence and examples, on important one is that there is always a lift in the following commutative diagram:

$$\begin{array}{ccccc} & & P & & \\ & \swarrow \exists! & \downarrow g & & \\ M & \xrightarrow{f} & N & \xrightarrow{0} & 0 \end{array}$$

where f is surjective. Naturally, free and stably free module are projective. A more nuanced example is that all $M_n(k)$ -modules (with $n > 1$) are projective but not free (which can be deduced using

Artin-Wedderburn). A more canonical set of examples comes from vector bundles. Take $R = C^0(X)$ to be the set of continuous real valued function on a compact topological space X , and let $\Psi : E \rightarrow X$ be a vector-bundle. Then the set $\Gamma(E)$ of continuous sections naturally forms a projective R -module. Let us show this (see [1] for the algebraic geometry perspective of the equivalence between vector bundle and locally free sheaves). The trivial bundle is $\mathbb{R}^n \times X \rightarrow X$ and we get $\Gamma(\mathbb{R}^n \times X) = R^n$. Now assume $\Psi : E \rightarrow X$ is a non-trivial vector-bundles (up to vector-bundle isomorphism). Then if $\Gamma(E)$ were free, there would be sections $\{s_1, \dots, s_n\}$ that would form a basis $f : T^n \rightarrow E$ which would give an isomorphism $\Gamma(T^n) = \Gamma(E)$. As the kernel and cokernel of bundles of f have no nonzero sections, they must vanish, and so f would be an isomorphism, and so $\Psi : E \rightarrow X$ would in fact be trivial.

If X is compact, then the famous *Swan's Theorem* gives an equivalence between the category of projective modules over $C^0(X)$ and the category of vector bundles over X , showing that vector bundles are the geometric interpretation of projective modules, which motivates similar discussions over sheaves of module for a scheme.

Proposition 1.1.3: Finitely Generated Projective Modules and Idempotents

Let P be a finitely generated projective R -module. Then P corresponds to an idempotent element in $M_n(R)$ for appropriate n .

Proof :

If P is finitely generated, then $P \oplus Q \cong R^n$, meaning there is an idempotent element in R which can be associated to an element of $M_n(R)$. To find this element, we can take the natural projection-inclusion

$$R^n \rightarrow P \rightarrow R^n$$

This map is an idempotent in $e \in M_n(R)$ (when representing this composition as a matrix under a choice of basis). Labeling this composition e , we see that $e(R^n) = P$.

If P is a projective R -module over a PID, then it is in fact free (as given by the structure theorem for modules over PID¹). Another famous class of modules where all finitely generated projectives are free comes from Quillen and Suslin stating that finitely generated modules over a polynomial ring (or a Laurent polynomial ring) over a PID is free². This then gives the result mentioned earlier about group rings over finite abelian groups, for $\mathbb{Z}[G]$ is the Laurent polynomial ring $\mathbb{Z}[x, x^{-1}, \dots, z, z^{-1}]$. On the other hand, if G is *not* abelian (but has the conditions of being torsion-free and nilpotent), then any projective $\mathbb{Z}[G]$ -module P is stably free but not free, in particular that:

$$P \oplus \mathbb{Z}[G] \cong (\mathbb{Z}[G])^2$$

As of writing this book, it is unknown if the condition “nilpotent” can be dropped and produce a similar (or the same) result.

Another important class of finitely generated projective R -module that are always free are those over *local rings*. For informative purposes, the result can in fact be extended to infinitely generated projective R -modules over local rings (Kaplansky gave such a proof). The following is given as a reminder:

¹Note this even works for infinitely generated projective modules, something briefly outlined in some exercises in [2]

²This result appears in [1] where it is used to show that all locally free $\mathcal{O}_X$ -modules on affine space must be free

Proposition 1.1.4: Projective Over Local Ring, then Free

Let R be a local ring. Then every finitely generated projective R -module is free, where

$$P \cong R^n \quad p = \dim_{R/\mathfrak{m}}(P/\mathfrak{m}P)$$

Proof :

see [2, chapter 20.3]

Proposition 1.1.5: Free At Stalks

Let R be a commutative ring, $\mathfrak{p}$ a prime ideal, and P a finitely generated projective R -module. Then the localization $P_{\mathfrak{p}}$ is isomorphic to $(R_{\mathfrak{p}})^n$ for some $n \geq 0$. Furthermore, there exists some $s \in R \setminus \mathfrak{p}$ where the localization away from s is free:

$$P \left[\frac{1}{s} \right] \cong R \left[\frac{1}{s} \right]^n$$

And we further get that $P_{\mathfrak{p}'} \cong (R_{\mathfrak{p}'})^n$ for every prime ideal $\mathfrak{p}'$ of R not containing s

Proof :

As P is projective, let $P \oplus Q = R^m$. Then $P_{\mathfrak{p}} \oplus Q_{\mathfrak{p}} = (R_{\mathfrak{p}})^m$, hence $P_{\mathfrak{p}}$ is a finitely generated projective $R_{\mathfrak{p}}$ -module. As $R_{\mathfrak{p}}$ is local, $P_{\mathfrak{p}}$ is in fact free. Now, every element of $P_{\mathfrak{p}}$ is of the form p/s , for $p \in P$ and $s \in R \setminus \mathfrak{p}$. By choosing a basis for $P_{\mathfrak{p}}$, we may clear the denominators to get an R -module homomorphism $\varphi : R^n \rightarrow P$ that become an isomorphism when localizing at $\mathfrak{p}$.

Now, as $\text{coker}(\varphi)$ is a finitely generated R -module which vanishes upon localization, it must be annihilated by some $s \in R \setminus \mathfrak{p}$. Given this s , we get a surjective map:

$$\varphi \left[\frac{1}{s} \right] : \left(R \left[\frac{1}{s} \right] \right)^n \rightarrow P \left[\frac{1}{s} \right]$$

As $P \left[\frac{1}{s} \right]$ is projective (as localization preserves projectiveness), $\left(R \left[\frac{1}{s} \right] \right)^n$ is isomorphic to the direct sum of $P \left[\frac{1}{s} \right]$ and a finitely generated $R \left[\frac{1}{s} \right]$ -module M (where $M_{\mathfrak{p}} = 0$). As M is annihilated by some $t \in R \setminus \mathfrak{p}$, we get:

$$\varphi \left[\frac{1}{st} \right] : \left(R \left[\frac{1}{st} \right] \right)^n \xrightarrow{\cong} P \left[\frac{1}{st} \right]$$

completing the proof.

We should next define the “vector bundle” structure that we can put on modules. To that end, we will have to assume some algebraic geometry, nothing more than some knowledge on spectra (see [2, chapter 21.7] or [1, chapter 2.1]). Let $\text{Spec } R$ be the collection of prime ideals of R , and define the Zariski topology with basic open sets given by:

$$D(s) = \{\mathfrak{p} \in \text{Spec } R \mid s \notin \mathfrak{p}\} \cong \text{Spec } R \left[\frac{1}{s} \right]$$

Definition 1.1.4: Rank At Prime Ideal

Let R be a commutative ring and M be a finitely generated R -module. Then the *rank* at $\mathfrak{p}$ is:

$$\text{rank}_{\mathfrak{p}}(M) = \dim_{\kappa(\mathfrak{p})} M \otimes_R \kappa(\mathfrak{p})$$

Note that $M_{\mathfrak{p}}/\mathfrak{p}M_{\mathfrak{p}} \cong \kappa(\mathfrak{p})^{\text{rank}_{\mathfrak{p}}(M)}$, hence $\text{rank}_{\mathfrak{p}}(M)$ is the minimal number of generators for $M_{\mathfrak{p}}$. If P is a finitely generated projective R -module, then $\text{rank}(P) : \mathfrak{p} \mapsto \text{rank}_{\mathfrak{p}}(P)$ is a continuous function between the $\text{Spec } R$ with the Zariski topology and $\mathbb{N}$ with the discrete topology.

If $\text{Spec } R$ is connected, then $\text{rank}(P) : \text{Spec } R \rightarrow \mathbb{N}$ must always be a constant function, hence every finitely generated projective R -module has constant rank. In the simplest case, R is an integral domain, then the rank is constant with:

$$\text{rank}(P) = \dim_{\text{Frac}(R)}(P \otimes_R \text{Frac}(R))$$

Note that if P is of constant rank, then it is finitely generated³. If M is not projective, then $\text{rank}(M)$ need not be continuous: consider $R = \mathbb{Z}$ with $M = \mathbb{Z}/p\mathbb{Z}$.

We conclude this section with an important observation about the “lack” of importance of higher dimensions. For the following, recall that M, M' are stably isomorphic if $M \oplus R^m \cong M' \oplus R^m$ for some $m \geq 0$.

Theorem 1.1.1: Base Serre Cancellation

Let R be a commutative Noetherian ring with $\text{kdim}(R) = d$, and let P be a projective R -module of constant rank $n > d$. Then:

1. $P \cong P_0 \oplus R^{n-d}$ for some projective R -module P_0 of constant rank d
2. If P is stably isomorphic to P' , then $P \cong P'$
3. For all M, M' , if $P \oplus M$ is stably isomorphic to M' , then

$$P \oplus M \cong M'$$

The idea is that any extra structure when considering higher dimension vector bundles only adds “trivial” information, making the information from vector-bundles ties to the local information of the base space.

Proof :

see H. Bass, Algebraic K-theory, Benjamin.

1.1.1 *Milnor Square

The following material is going to be more useful in section 2.5 when looking over Mayer-Vietoris sequences, however it appears when studying the Picard group of a ring. The idea is that the Milnor square gives us a way of gluing together different parts of a ring akin to the Mayer-Vietoris sequence from algebraic topology (see [5]). Recall from algebraic topology that this construction was express done to be computationally rather simple. We “algebras” this technique by looking at a particularly

³Prove it for the case of rank 1, then for the rank r case show that $\Lambda^r(P)$ is finitely generated

nice way to “patch” two rings in such a way that will allow for some simple computations of various invariance (ex. Picard group, K -groups, etc)

Let R be a ring, $I \subseteq R$ an ideal, $\varphi : R \rightarrow S$ a ring homomorphism that maps I isomorphically to an ideal in S , which we shall also label I . Then R is the pullback of S and R/I in the following diagram:

$$\begin{array}{ccc} R & \longrightarrow & S \\ \downarrow & & \downarrow \\ R/I & \longrightarrow & S/I \end{array}$$

This square is called the *Milnor square*. In algebraic geometry, this represents R as the fibered product of S and R/I over S/I .

We can patch modules given a Milnor square. Let M_1 be an S -module and M_2 be an R/I -module. The Milnor square defines the natural S/I -module isomorphism:

$$g : M_2 \otimes_{R/I} S/I \cong M_1/IM_1$$

Then M can be thought of as a triple (M_1, g, M_2) and is a subset of $M_1 \times M_2$ given by the kernel of the R -module homomorphism:

$$M_1 \times M_2 \rightarrow M_1/IM_1 \quad (m_1, m_2) \mapsto \overline{m}_1 - g(\overline{f}(m_2))$$

Example 1.3: Milnor Patching A Module

Take S^n and $(R/I)^n$ with $1 \in \text{GL}_n(S/I)$. Consider $\overline{f} : R/I \rightarrow S/I$ Then:

$$\left\{ (\vec{s}, r + I) \mid \vec{s} = \overline{f}(r + I) \right\}$$

Hence, we have patched together two vector-spaces of the same dimension over different rings that can glue over R/I , or geometrically we have two trivial bundles over R/I and S and we glued those bundles to create a bundle over R .

The most important properties of Milnor patching that we shall use is summarized in the following:

Theorem 1.1.2: Milnor patching

Let

$$\begin{array}{ccc} R & \longrightarrow & S \\ \downarrow & & \downarrow \\ R/I & \longrightarrow & S/I \end{array}$$

be a commutative diagram where $I \subseteq R$ is identified with its image in S and also represented as I (i.e. a Milnor square). Then:

1. If P is obtained by patching together a finitely generated projective S -module P_1 and a finitely generated projective R/I -module P_2 , then P is a finitely generated projective R -module.
2. $P \otimes_R S \cong P_1$ and $P/IP \cong P_2$.
3. Every finitely generated projective R -module arises in this way.
4. If P is obtained by patching free modules along $g \in GL_n(S/I)$ and Q is obtained by patching free modules along g^{-1} , then $P \oplus Q \cong R^{2n}$.

Proof :

1. Let's start with a Milnor square, which is a pullback diagram:

$$\begin{array}{ccc} R & \rightarrow & S \\ \downarrow & & \downarrow \\ R/I & \rightarrow & S/I \end{array}$$

Suppose we have a finitely generated projective S -module P_1 and a finitely generated projective R/I -module P_2 , along with an isomorphism $g : P_1 \otimes_S S/I \cong P_2/IP_2$. Let

$$P = \{(x, y) \in P_1 \times P_2 \mid \bar{x} = g(\bar{f}(y))\}$$

To show P is finitely generated, Since P_1 and P_2 is a finitely generated, there exists a surjection $S^n \rightarrow P_1$ for some n and a surjection $(R/I)^m \rightarrow P_2$ for some m . Then it is certainly the case that we can construct a surjection $R^{m+n} \rightarrow P$, showing that P is finitely generated.

For showing it is projective, since P_1 is projective over S , there exists an S -module Q_1 such that $P_1 \oplus Q_1 \cong S^n$ for some n . Similarly, there exists an R/I -module Q_2 such that $P_2 \oplus Q_2 \cong (R/I)^m$ for some m .

The isomorphism $g : P_1 \otimes_S S/I \cong P_2/IP_2$ extends $\tilde{g} : S^n \otimes_S S/I \cong (R/I)^m/I(R/I)^m$, namely it is the isomorphism induced by taking $Q_1 \otimes_S S/I$ and Q_2/IQ_2 . Now, define F by:

$$F = \{(x, y) \in S^n \times (R/I)^m \mid \bar{x} = g(\bar{f}(y))\}$$

By the properties of the Milnor square, F is a free R -module. The inclusion maps $P_1 \hookrightarrow S^n$ and $P_2 \hookrightarrow (R/I)^m$ induce an inclusion $P \hookrightarrow F$. Moreover, the projections $S^n \rightarrow P_1$ and $(R/I)^m \rightarrow P_2$ induce a projection $F \rightarrow P$ such that the composition $P \rightarrow F \rightarrow P$ is the identity on P . But then P is projective,

2. This follows by our above construction

3. If M is any R -module, the Milnor square gives a natural map from M to the R -module M' by patching $M_1 = M \otimes_R S$ and $M_2 = M \otimes_R (R/I) = M/IM$ along the natural isomorphism

$$(M/IM) \otimes_{R/I} (S/I) \cong M \otimes_R (S/I) \cong (M \otimes_R S)/I(M \otimes_R S)$$

Now, tensor M with $0 \rightarrow R \rightarrow (R/I) \oplus S \rightarrow S/I \rightarrow 0$. This gives an exact sequence (recall that tensoring is right-exact, but not necessarily left-exact).

$$\mathrm{Tor}_1^R(M, S/I) \rightarrow M \rightarrow M' \rightarrow 0$$

so in general M' is just a quotient of M . However, if M is projective, the Tor term is zero and $M \cong M'$. Thus every projective R -module may be obtained by patching.

1.2 Picard Group of a Commutative Ring

Let us now tackle the connection between projective modules and vector bundles more closely. To fully understand the interplay between the algebraic and the geometric, the following theorem must be quoted:

Theorem 1.2.1: Serre-Swan's Theorem

1. **(Serre's Theorem)** Let R be a commutative Noetherian ring. Then the category of finitely generated projective R -modules is equivalent to the category of algebraic vector bundles^a on $\mathrm{Spec} R$
2. **(Swan's Theorem)** Let X be a compact Hausdorff space. Then the category of finitely generated projective R -modules over the continuous functions $C(X)$ is equivalent to the category of finite-rank vector bundles on X , the equivalence being established by sending vector bundles its module of continuous section.

^alocally free sheaves of structure sheaf-modules of constant finite rank

Proof :

See Serre's paper and Swan's paper.

Due to these connections, we have the following definition:

Definition 1.2.1: Algebraic Line Bundle

Let R be a commutative ring. Then an *algebraic line bundle* L over R is a finitely generated projective R -module of constant rank 1.

In this section, we shall just say *line bundle*. As line bundles are locally 1 dimensional, it is easy to verify that the tensor product of line bundles is again a line bundle (hint: $\dim(V \otimes W) = \dim(V) \cdot \dim(W)$). Furthermore, it should be immediate to the current reader that $L \otimes_R R \cong L$ for any line bundle (even projective module). Then $\otimes_R$ forms a commutative monoid. If we show there is an inverse, we have a group, and indeed:

Lemma 1.2.1: Inverse Line Bundle

Let L be a line bundle and $L^\vee = \text{Hom}_R(L, R)$. Then

$$L \otimes_R L^\vee \cong R$$

Proof :

First, $L^\vee$ is line bundle as $\text{rank}(L) = \text{rank}(L^\vee)$. Next, take:

$$f \otimes_R \ell \mapsto f(\ell)$$

If $L = R$, this is certainly an isomorphism. Hence in the general case, localizing at $\mathfrak{p}$ we get:

$$(L^\vee \otimes_R L)_\mathfrak{p} \cong L^\vee_\mathfrak{p} \otimes_{R_\mathfrak{p}} L_\mathfrak{p} \rightarrow R_\mathfrak{p}$$

which is then certainly an isomorphism. As being an isomorphism is a local property, we get the global isomorphism, completing the proof.

Definition 1.2.2: Picard Group

Let R be a commutative ring. Then the *picard group* of R is the set of isomorphism classes of line bundles on R with the operation being the tensor product $\otimes_R$ and inverses $L^{-1} = L^\vee$.

Proposition 1.2.1: Picard Group Homomorphism

Let $R \rightarrow S$ be a ring homomorphism between commutative rings. Then $\text{Pic}(R) \rightarrow \text{Pic}(S)$ is a group homomorphism mapping $L \mapsto L \otimes_R S$

Proof :

Certainly $L \otimes_R S$ is a line bundle over S . Then it is an exercise in algebra to see that:

$$(L \otimes_R M) \otimes_R S \cong (L \otimes_R S) \otimes_R (M \otimes_R S)$$

showing that $\otimes_R S$ is a group-homomorphism, as we sought to show.

We prove the following representation of line bundles for later:

Lemma 1.2.2: Representation of Line Bundle

Let L be a line bundle over R . Then

$$\text{End}_R(L) \cong R$$

Proof :

Multiplication by elements of r give a map $R \rightarrow \text{End}_R(L)$. This map is locally an isomorphism, and hence an isomorphism, completing the proof.

Example 1.4: Picard Groups of Commutative Rings

1. Let $R = k$. Then as k is a field, every projective k -module is free. But then any projective k module of rank 1 is a dimension 1 vector-space, namely it is isomorphic to k . Hence:

$$\text{Pic}(k) = 0$$

2. Similarly, as every module over a PID is free, if $R = \mathbb{Z}$, then:

$$\text{Pic}(\mathbb{Z}) = 0$$

3. Let R be a Dedekind domain (usually a number ring). We shall heavily be relying on the algebra presented in [2, chapter 22.6]. First, recall that each ideal $I \subseteq R$ is a projective R -module, and it is certainly locally of rank 1 (every ideal is generated by at most 2 elements, and localizing at the appropriate prime makes the ideal principal). Next, recall that any torsion-free R -module of rank n is isomorphic to:

$$M \cong R^{n-1} \oplus I$$

for an ideal I that can be replaced by $J = (a)I$ for $a \in R$. As all projective modules are torsion free, and we are looking at rank 1 modules, we see that the line bundles on R are in correspondence to the ideals of R . As principal ideals are isomorphic to R , we have that:

$$\text{Pic}(R) \cong \text{Cl}(R)$$

4. (If you know some divisor theory) Building off the work from the last point, If X is a nonsingular affine variety over an algebraically closed field $\bar{k}$ then $\text{Pic}(k[X])$ (the Picard group of its coordinate ring) is isomorphic to the divisor class group of X .
5. it was proven by Quillen and Suslin that every projective module over polynomial ring is free. Thus, the picard group of $k[x_1, \dots, x_n]$ is always trivial:

$$\text{Pic}(k[x_1, \dots, x_n]) = 0$$

6. (foreshadowing for elliptic curve theory). An elliptic curve is a smooth algebraic curve (over an algebraically closed field $\bar{k}$) whose genus is 1^a. We shall see that there is a natural subgroup of the Picard group, Pic^0 , and that if E is an elliptic curve associated to $R = \bar{k}[x, y, z]/I$, then:

$$\text{Pic}^0(R) \cong R$$

Which is a rather interesting idempotence! See cite:AlgGeo for more details.

^aEvery nonsingular curve maps onto a compact Riemann surface of genus g

We shall next define a natural notion known as the *determinant line bundle*. Its key feature of interest to us is that all finitely generated projective R -modules are determined by their rank and determinant. First, recall the following properties of the exterior product:

Proposition 1.2.2: Properties of The Exterior Product

Let R be a commutative ring and M an R -module. Then:

1. $\Lambda^k(R^n)$ is a free R -module of rank $\binom{n}{k}$ generated by $e_{i_1} \wedge \cdots \wedge e_{i_n}$ where $i_1 \leq \cdots \leq i_n$.
Note that $\Lambda^n(R^n) \cong R$
2. If $R \rightarrow S$ is a ring map, there is an isomorphism:

$$(\Lambda^k M) \otimes_R S \cong \Lambda^k(M \otimes_R S)$$

3. There is a natural isomorphism:

$$\bigwedge^k (P \oplus Q) \cong \bigoplus_{i=0}^k \left(\bigwedge^i P \right) \otimes \left(\bigwedge^{k-i} Q \right)$$

4. If P is projective, $\Lambda^k(P)$ is projective (as it is locally free).

Proof :

exercise, or see [2, chapter 15.4]

Recall that $\Lambda^n(M)$ for a n -rank R -module is generated by a single element, and by moving all the coefficients to the front we get the determinant. This motivates the following terminology:

Definition 1.2.3: Determinant Bundle

Let P be a finitely generated projective R -module of constant rank n . Then $\Lambda^n(P)$ is called the *determinant line bundle*, and is denoted $\det(P)$.

If P is not constant rank, we define it component-wise. If $g : P \rightarrow P$ is an endomorphism, $\det(-)$ is a functor $\deg(g) : \det(P) \rightarrow \det(P)$, and by lemma 1.2.2 we can think of $\deg(g) \in R$.

We shall state, but not prove, the following useful classification result:

Proposition 1.2.3: Classification of Projective Module Over 1-dimensional Ring

Let R be a commutative Noetherian 1-dimensional ring. Then all finitely generated projective R -modules are completely classified by their rank and determinant. In particular, every finitely generated projective R -module P of rank ≥ 1 is isomorphic to $L \oplus R^f$, where $L = \det(P)$ and $f = \text{rank}(P) - 1$.

Proof :

hint: Show that R is the only stably free-line bundle.

Let us next focus on line bundles over integral domains. We shall see that all line bundles on integral domains are given by *invertible ideal*:

Proposition 1.2.4: Classification of Projective Modules over Integral Domains

Let R be a commutative integral domain. L is a line bundle if and only if it is (isomorphic to) an invertible ideal. If I and J are fractional ideals and I is invertible, then $I \otimes_R J \cong IJ$. Furthermore, there is an exact sequence of abelian groups:

$$1 \rightarrow R^\times \rightarrow F^\times \xrightarrow{\text{Div}} J_K(R) \rightarrow \text{Pic}(R) \rightarrow 0.$$

where $J_K(R)$ is the ideal group.

Proof :

If I and J are invertible ideals such that $IJ \subseteq R$, then we can interpret elements of J as homomorphisms $I \rightarrow R$. If $IJ = R$, then we can find $x_i \in I$ and $y_i \in J$ so that $x_1y_1 + \cdots + x_ny_n = 1$. The $\{x_i\}$ assemble to give a map $R^n \rightarrow I$ and the $\{y_i\}$ assemble to give a map $I \rightarrow R^n$. The composite $I \rightarrow R^n \rightarrow I$ is the identity, because it sends $r \in I$ to $\sum x_iy_ir = r$. Thus I is a summand of R^n , i.e., I is a finitely generated projective module.

As R is an integral domain and $I \subseteq F$, $\text{rank}(I)$ is the constant $\dim_F(I \otimes_R F) = \dim_F(F) = 1$. Hence I is a line bundle.

This construction gives a set map $J_K(R) \rightarrow \text{Pic}(R)$; to show that it is a group homomorphism, it suffices to show that $I \otimes_R J \cong IJ$ for invertible ideals. Suppose that I is a submodule of F which is also a line bundle over R . As I is projective, $I \otimes_R -$ is an exact functor. Thus if J is an R -submodule of F , then $I \otimes_R J$ is a submodule of $I \otimes_R F$. The map $I \otimes_R F \rightarrow F$ given by multiplication in F is an isomorphism because I is locally free and F is a field. Therefore the composite

$$I \otimes_R J \subseteq I \otimes_R F \xrightarrow{\text{multiply}} F$$

sends $\sum x_i \otimes y_i$ to $\sum x_iy_i$. Hence $I \otimes_R J$ is isomorphic to its image $IJ \subseteq F$. This proves the third assertion.

The kernel of $J_K(R) \rightarrow \text{Pic}(R)$ is the set of invertible ideals I having an isomorphism $I \cong R$. If $f \in I$ corresponds to $1 \in R$ under such an isomorphism, then $I = fR = \div(f)$. This proves exactness of the sequence at $J_K(R)$.

Clearly the units $R^\times$ of R inject into $F^\times$. If $f \in F^\times$, then $fR = R$ if and only if $f \in R$ and f is in no proper ideal, i.e., if and only if $f \in R^\times$. This proves exactness at $R^\times$ and $F^\times$.

Finally, we have to show that every line bundle L is isomorphic to an invertible ideal of R . Since $\text{rank}(L) = 1$, there is an isomorphism $L \otimes_R F \cong F$. This gives an injection $L \cong L \otimes_R R \subseteq L \otimes_R F \cong F$, i.e., an isomorphism of L with an R -submodule I of F . Since L is finitely generated, I is a fractional ideal. Choosing an isomorphism $\tilde{L} \cong J$, Lemma 3.1 yields

$$R \cong L \otimes_R \tilde{L} \cong I \otimes_R J \cong IJ.$$

Hence $IJ = fR$ for some $f \in F^\times$, and $I(f^{-1}J) = R$, so I is invertible.

There is further discussion different interesting divisor groups, class groups, and their relation to the Picard group, but we leave that material to cite:algGeo.

1.3 Vector Bundles on Topological Spaces

Many of the results covered apply to paracompact topological spaces (every open cover has a locally finite refinement). We require the paracompact condition so that many of the constructions we shall want to have will work, namely we have good behaviour of sheaves and sufficient conditions for homotopy theory and differential topology. One interesting phenomena that is not present when working over spectra is *Bott Periodicity* which we endeavor to show in this section. We shall assume a good understanding of the theory of vector bundles; the necessary results can be seen in [8, chapter 6] and [5, chapter 5]. Some important results that are worth recalling are:

1. Sub-bundle shave compliments: if $E \rightarrow X$ is a bundle (on a paracompact topological space) and F is a subbundle, then there exists a bundle $F^\perp$ such that $E = F \oplus F^\perp$.
2. Every vector bundle over a compact space is the Whitney direct summand of a trivial bundle.
3. the Tensor product is well-defined: Given a vector bundle $E \rightarrow X$ and $F \rightarrow X$, there exists a vector-bundle $E \otimes F \rightarrow X$ whose fibers over x are $E_x \otimes F_x$. From this, we get our usual more general bundles such as the *determinant bundle* $\det(E) = \Lambda^n E$ (given E is n -dimensional).
4. If the transition maps are in O_n , U_n , or Sp_n , we saw that the vector bundle has a *structure group* O_n , U_n , or Sp_n respectively. We can convert any bundle to have a structure group as any bundle has a (Riemann or Hermitian) metric.
5. If $f, g : X \rightarrow Y$ are homotopic and Y is paracompact and $E \rightarrow X$ is a vector-bundle, then:

$$f^*E \cong g^*E$$

As a consequence, if Y is contractible then every vector bundle on Y is trivial.

One important class of vector bundles are those on spheres S^n . Let us take a moment to understand these as they form an important class of examples and form a surprising periodicity in their higher-order homotopy known as *Bott Periodicity*. The main idea behind looking at the bundles on spheres is by looking at the behaviour of bundle after a suspension. (as S^n is the suspension of S^{n-1}).

Proposition 1.3.1: Vector bundles and Suspension

Let SX denote the suspension of a paracompact space X . Then given a choice of base-point $x \in X$, there is a 1-1 correspondence between the set $\mathbf{VB}_n(SX)$ of isomorphism classes of n -dimensional (resp. real, complex, or quaternionic) vector bundles over SX and the respective set of based homotopy classes of maps

$$[X, O_n]_*, \quad [X, U_n]_*, \quad \text{or} \quad [X, Sp_n]_*$$

from X to the orthogonal group O_n , unitary group U_n , or symplectic group Sp_n .

Proof :

The proof is a complicated proof, the following is a sketch given from [3]. From [5], we know that SX is the union of two contractible cones C_1 and C_2 whose intersection is X . As every vector bundle on the cones C_i is trivial as it is contractible, and every vector bundle on SX is obtained

from an isomorphism of trivial bundles over X via the clutching construction (see [3]; it is the Mayer-Vietoris sequence for this suspension construction). Such an isomorphism is given by a homotopy class of maps from X to GL_n , or equivalently, to the appropriate deformation retract $(O_n, U_n, \text{ or } Sp_n)$ of GL_n . Then the indeterminacy in the resulting map from $[X, GL_n]$ to classes of vector bundles is eliminated by insisting that the base point of X map to $1 \in GL_n$.

Now, from [7] or [5], we know the computations of the homotopy groups classical Lie groups, which we now take advantage of to find the isomorphism classes of vector bundles on spheres

Example 1.5: Vector Bundles On Spheres

1. Let $X = S^1$, and recall that $|\pi_0(O_n)| = 2$ for real vector bundles of dimension n for all $n \geq 1$. This corresponds to the trivial bundle and the möbius bundle. The Whitney direct sum of these two shall give all non-trivial real vector bundles on S^1 .

As $|\pi_0(U_n)| = 1$ for all n , all complex vector bundles are trivial!

2. Let $X = S^2$. Then $\pi_1(O_1) = 0$, and so there is only the trivial real line bundle on S^1 . As $\pi_1(O_2) = \mathbb{Z}$, there are infinitely many 2-dimensional real line bundles of dimension 2 (indexed by the degree of the clutching map mentioned in the above proof). As $\pi_1(O_n) = \mathbb{Z}/2\mathbb{Z}$ for $n \geq 2$, there is only two real line bundles for dimension $n \geq 3$.

For complex line bundles on S^3 , we get the same phenomena as $\pi_1(U_2) = \mathbb{Z}$

3. Recall from [7] that $\pi_2(G) = 0$ for all compact Lie groups. Thus, all vector bundles on S^3 are trivial!
4. From the above, we can conclude that all (real and complex) line bundles on S^4 trivial. as:

$$\begin{array}{lll} \pi_3(O_n) = 3 & \pi_3(U_n) = \mathbb{Z} & \pi_3(Sp_n) = \mathbb{Z} \\ \text{for } n \geq 5 & \text{for } n \geq 2 & \text{for } n \geq 1 \end{array}$$

We get the corresponding infinite family of n -dimensional Real/Complex/Quaternion vector bundles.

With some examples under our belt, let us look into broader classification of vector bundles. This is a major result, and so we shall simply outline the idea for the readers awareness. This is something special about vector bundles on paracompact spaces as opposed to the exploration we have done on projective modules, namely due to the restriction to the vector spaces being over $\mathbb{R}$ or $\mathbb{C}$, and that the base space has a nicer topology than the Zariski topology on $\text{Spec } R$. Let V be a finite dimensional vector space and let $\text{Grass}_k(V)$ be all k -dimensional subspaces of V ; i.e. the *Grassmannian manifold* of all sub-spaces of V (see [8] for a proof that it is a manifold). The *infinite Grassmannian* Grass_n is the union all $\text{Grass}_n(V)$ as V ranges over all finite dimensional subspace of a fixed infinite dimensional vector space (usually $\mathbb{R}^\infty$, $\mathbb{C}^\infty$, or $\mathbb{H}^\infty$). With a bit of work, we see that it is an infinite-dimensional CW complex (it is the limit of CW complexes respecting the appropriate directed mappings). This are rather natural, for example if taking $n = 1$, we get that Grass_1 is $\mathbb{RP}^\infty$, $\mathbb{CP}^\infty$, or $\mathbb{HP}^\infty$ respectively.

From this, we get the following classification result:

Theorem 1.3.1: Classifying Vector Bundles

Let X be a paracompact space. Then $\mathbf{VB}_n(X)$ is in correspondence with the set of homotopy classes of maps $X \rightarrow \text{Grass}_n$, denoted $[X, \text{Grass}_n]$. In other words:

$$\mathbf{VB}_n(X) \cong [X, \text{Grass}_n]$$

In particular, if $E \rightarrow X$ is an n -dimensional vector bundle, it is isomorphic to $f^*(E_n)$ for some $f : X \rightarrow \text{Grass}_n$ where E determines f up to homotopy.

Proof :

The proof can be found in *Fibre Bundles* by D. Husemoller.

We may take this a step further: as we saw we can put a natural structure group on vector bundles (i.e. O_n , U_n or Sp_n depending on whether the vector bundle is a real, complex, or quaternion bundle respectively). Hence, the infinite Grassmannian is called “classifying space” of O_n , U_n or Sp_n . For this reason, the infinite Grassmannian is denoted BO_n , BU_n , or BSp_n respectively.

2

K_0 Group

We saw the Picard group allowed us to treat the collection of (isomorphism classes of) line bundles of constant rank one as a group under $\otimes$. We shall now be looking at a similar situation with the direct sum.

2.1 Algebraic Build-up

Let M be an abelian monoid. Then we universally augment it into an abelian group, $M^{-1}M$ through the usual free-quotient construction. In particular, take $\mathcal{F}(M \times M) / ([m+n] - [m] - [n])$. This is called the *Grothendieck Completion*. A *Cancellation monoid* is a monoid where $m+a = n+a$ if and only if $m = n$.

Proposition 2.1.1: Properties of Monoid Completion

Let M be a monoid, $M^{-1}M$ be its completion. Then:

1. Every element of $M^{-1}M$ can be written as $[m] - [n]$ for $m, n \in M$
2. For each $[m], [n] \in M^{-1}M$, $[m] = [n]$ if and only if there exists a p such that $m+p = n+p$.
3. The map $M \times M \rightarrow M^{-1}M$ where $(m, n) \mapsto [m] - [n]$ is surjective
4. M injects into $M^{-1}M$ if and only if M is a cancellation monoid.
5. The Grothendieck Completion is a functor that is left adjoint to the forgetful functor from the category of abelian group to the category of abelian monoids.

Proof :

these are all simple exercises, but you can take a look at [2, chapter 15].

If we augment a monoid with a 2nd operation that satisfies the ring axioms, then we get a *semiring*. The semiring $\mathbb{N}$ is the initial semiring in this category. Then the Grothendieck functor can also be seen as a functor from semirings to rings.

When we shall be studying $K_0(R)$ groups, we shall find that the free R -modules play an important role. These elements are important enough so that we consider the following special type of subgroup of a monoid

Definition 2.1.1: Cofinal Monoid

Let M be a monoid. Then $L \subseteq M$ is called *cofinal* if for every $m \in M$, there exists a $m' \in M$ such that $m + m' \in L$.

The terminology *cofinal* comes from order theory, where a subset $A \subseteq B$ of an preorder $(B, \leq)$ is cofinal if for every $b \in B$, there is an $a \in A$ such that $b \leq a$.

Lemma 2.1.1: Cofinal Monoids

Let $L \subseteq m$ be a cofinal monoid. Then:

1. $L^{-1}L \subseteq M^{-1}M$ is a subgroup
2. Every element $[m] \in M^{-1}M$ can be written as $[m'] - [\ell]$ for $m' \in M$ and $\ell \in L$.
3. If $[m] = [m'] \in M^{-1}M$, then $m + \ell = m' + \ell$ for some $\ell \in L$.

Proof :

this is a matter of checking definitions

Example 2.1: Grothendieck Completion

1. some good examples in the book!

2.2 K_0 of a ring

Definition 2.2.1: First K-ring: $K_0(R)$

Let R be a [not necessarily commutative] ring, $P(R)$ the collection of (isomorphism classes of) finitely generated projective R -modules. Then combining it with $\oplus$ and appending an identity element 0, we get an abelian monoid. Its Grothendieck completion is the group $K_0(R)$.

When R is commutative, then along with $\otimes_R$, this forms a ring.

We restrict to finitely generated modules simply because the usual behaviour of infinitely generated modules trivializes the structure, for example if P is any finitely generated projective R -module then, then if we had to consider the infinitely generated module R^∞ we get:

$$P \oplus R^\infty \cong R^\infty$$

Therefore $[P] = 0$ and $K_0(R) = 0$, and so we naturally stick to the finitely generated case.

If R is commutative, then $K_0(R)$ is commutative with identity $1 = [R]$. This is indeed the identity, for $\oplus$ and $\otimes$ distribute, and $P \otimes_R R \cong P$. The canonical trivial example is when $R = k$ (or any division ring). Then $K_0(k) \cong \mathbb{Z}$. Similarity, if R is a local ring or a PID, then $K_0(R) \cong \mathbb{Z}$ (as these are sufficient conditions for a projective module to be free). We shall give non-trivial example after we have developed some more theory to be able to identify which properties lead to structural results for $K_0(R)$.

It should be noted that $K_0(-)$ is a functor from **Ring** to **Ab** or from **CRing** to **CRing**, namely if $\varphi : R \rightarrow S$ is a ring homomorphism, then $\otimes_R S : P(R) \rightarrow P(S)$ is a monoid map which induces the map $K_0(R) \rightarrow K_0(S)$. If R, S are commutative, then $\otimes_R S : P(R) \rightarrow P(S)$ is a semiring map as:

$$(P \otimes_R Q) \otimes_R S \cong (P \otimes_R S) \otimes_R (Q \otimes_R S)$$

and hence we get a ring homomorphism. This shall become handy when trying to translate properties from known $K_0(R)$ groups to one's we are trying to study.

In $K_0(R)$, the free-modules play a particularity important role. First of all, they are *cofinal* in $P(R)$ (see definition 2.1.1). Then every element of $K_0(R)$ can be written as:

$$[P] - [R^n]$$

Furthermore, by lemma 2.1.1, we have $[P] = [Q]$ if and only if:

$$P \oplus R^n \cong Q \oplus R^m$$

that is, P, Q are stably isomorphic, and $[P] = [R^n]$ if P is stably free. If R satisfies the IBP, then $L \subseteq P(R)$ is isomorphic to $\mathbb{N}$. If R respect IBP, we get the following useful characterization of trivial K_0 groups:

Lemma 2.2.1: Ibp and Trivial K_0

Let $\mathbb{N} \rightarrow P(R)$ be the monoid map sending $n \rightarrow R^n$ inducing the group homomorphism $\mathbb{Z} \rightarrow K_0(R)$. Then:

1. this map is injective if and only if R satisfies IBP
2. If R satisfies IBP, then

$K_0(R) \cong \mathbb{Z}$ if and only if every finitely generated projective R -module is stably free

Proof :

1. This is immediate by IBP
2. $P \oplus R^n \cong R^m$, and so each $[P] = [R^m]$ for some n .

Example 2.2: K_0 of some rings

1. As mentioned before, fields/division rings, PIDs, and local rings all have $K_0(R) = \mathbb{Z}$

2. If $R = R_1 \times R_2$, then by [2, chapter 23] we have the equivalence of categories:

$$(R \times S)\text{-Mod} \cong R\text{-Mod} \times S\text{-Mod}$$

Thus we get $P(R) = P(R_1) \times P(R_2)$, and its completion gives $K_0(R) = K_0(R_1) \times K_0(R_2)$. We may thus compute $K_0(R)$ component-wise.

3. Let $R = M_n(F)$. Then R is a simple ring, and so every $M_n(F)$ -module is projective. Recall that an $M_n(F)$ -module is some direct sums of F , and the *length* of an $M_n(F)$ -module is the number of copies. Then the length gives a natural map:

$$K_0(M_n(F)) \cong \mathbb{Z}$$

Note that $M_n(F)$ has length n , and so the subgroup generated by the free modules have index n , so the natural inclusion map $\mathbb{Z} \rightarrow K_0(M_n(F))$ does not split!

4. Let R be a semi-simple ring so that by Artin-Wedderburn:

$$R \cong M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$$

Thus, we get:

$$K_0(R) = \prod_i K_0(M_{n_i}(D_i)) \cong \mathbb{Z}^k$$

5. If R is a commutative ring, we have a natural map from R to a field, $R \rightarrow R/\mathfrak{m}$ for some maximal ideal $\mathfrak{m}$; call this field F . Then this induces a natural map $K_0(R) \rightarrow K(F)$, that is $K_0(R) \rightarrow \mathbb{Z}$ which maps $[R]$ to 1. Note that there doesn't always exist, such a map, we shall see an example where $K_0(R) \cong \mathbb{Q}$.
6. (If you know some algebraic geometry) Recall that a sheaf is *flasque* if all restriction maps are surjective. The key interesting property of Flasque sheaves is that they have trivial cohomology, $H^i(X, \mathcal{F}) = 0$ for $i > 0$ and $\mathcal{F}$ a flasque sheaf. We shall link the $K_0(R)$ group to (co)homological considerations, and will find the following definition to be the appropriate purely algebraic counter-part. We say a ring R is *flasque* if there is an R -bimodule M that is finitely generated and projective as a right R -module, such that there is a R -bimodule isomorphism:

$$\varphi : R \oplus M \xrightarrow{\sim} M$$

From this, we get that $K_0(R) = 0$. To see this, note that:

$$P \oplus (P \otimes_R M) \cong P \otimes_R (R \oplus M) \cong (P \otimes_R M)$$

Hence $[P] + [P \otimes_R M] = [P \otimes_R M]$, implying $[P] = [0]$. If R is flasque and the right R -module structure of M is R , then R is said to be an *infinite sum ring*. The isomorphism φ becomes $R^2 \cong R$. An example of such a ring is $\text{End}_R(R^\infty)$.

7. (There is an example on p.76 of the K-theory textbook I'm following on Von Neumann regular rings, which are rings where for every $x \in R$, there exists a $y \in R$ such that $xyx = x$. These are rings in which inverse need not exist, but they almost do, for example each field is Von Neumann regular rings because we can let $y = x^{-1}$. No integral domain is Von Neumann regular, but $M_n(F)$ is Von Neumann regular. There is a natural lattice structure that comes out of its idempotence that $K_0(R)$ is isomorphic to if R is a Von Neumann regular rings)

8. let $R = \varinjlim_i R_i$ where $\{R_i\}$ is a filtered system. Then every finitely generated projective R -module is of the form:

$$R \otimes_{R_i} P_i$$

where P_i is some finitely generated projective R_i -module. Any isomorphism:

$$R \otimes_{R_i} P_i \cong R \otimes_{R_i} P'_i$$

must set a finite set of generators to a finite set of generators, and so for j large enough this reduces to:

$$R_j \otimes_{R_i} P_i \cong R_j \otimes_{R_i} P'_i$$

But that means that $P(R)$ is then a filtered colimit of $P(R_i)$! We thus have the following wonderful result about colimits passing through K -groups:

$$K_0(R) = \varinjlim K_0(R_i)$$

If R is commutative, we can show that every R is the colimit of finitely generated subrings! As every finitely generated commutative ring is Noetherian with finite normalization (the integral closure is finitely generated over the ring), we may reduce studying $K_0(R)$ to the study of $K_0(R_i)$ where the R_i have nicer properties!

The following is very useful and allows for interesting geometric interpretations. For the following, we shall say that I is *complete* if every Cauchy sequence $\sum_i^\infty a_i$ where $a_i \in R/I^n$ converges to a unique element in I .

Theorem 2.2.1: K-group and Nilpotence

Let R be a ring and $I \subseteq R$ be nilpotent, or more generally complete (see definition above). Then:

$$K_0(R) \cong K_0(R/I)$$

And if R is commutative, then:

$$K_0(R) \cong K_0(R_{\text{red}})$$

Proof :

We'll show $P(R) \cong P(R/I)$, which shall complete the result. We first need to build-up some important result:

lemma 1: Let P_1, P_2 be finitely generated projective R -modules and $I \subseteq R$ a radical idea. Then $P_1 \cong_R P_2$ if and only if $P_1/IP_1 \cong_R P_2/IP_2$

Proof. Certainly $P_1 \cong_R P_2$ implies $P_1/IP_1 \cong_R P_2/IP_2$, so we do the converse. There is a natural map $P_1 \rightarrow P_1/IP_1$, and then there is an isomorphism $\varphi : P_1/IP_1 \rightarrow P_2/IP_2$. Finally, there is a natural projection $\pi : P_2 \rightarrow P_2/IP_2$:

$$\begin{array}{ccc} P_1 & & P_2 \\ \pi \downarrow & \searrow \varphi \circ \pi & \downarrow \pi \\ P_1/IP_1 & \xrightarrow{\varphi} & P_2/IP_2 \end{array}$$

As P_1 is projective, there is a natural lift making the following diagram commute:

$$\begin{array}{ccc} P_1 & \xrightarrow{\tilde{\varphi}} & P_2 \\ \pi \downarrow & & \downarrow \pi \\ P_1/IP_1 & \xrightarrow{\varphi} & P_2/IP_2 \end{array}$$

If we show $\tilde{\varphi}$ is an isomorphism, we're done. Note that as φ is an isomorphism, in particular it is surjective, then as I is radical it is a consequence of Nakayama's lemma that $\varphi : P_1 \rightarrow P_2$ is surjective (see exercises in [2, Chapter 20.3] or checkout [6]). Let's thus show injectivity. To that end, let us localize at each $\mathfrak{p} \subseteq R$ and consider:

$$\begin{array}{ccc} P_{1,\mathfrak{p}} & \xrightarrow{\tilde{\varphi}_{\mathfrak{p}}} & P_{2,\mathfrak{p}} \\ \pi \downarrow & & \downarrow \pi \\ P_{1,\mathfrak{p}}/I_{\mathfrak{p}}P_{1,\mathfrak{p}} & \xrightarrow{\varphi_{\mathfrak{p}}} & P_{2,\mathfrak{p}}/I_{\mathfrak{p}}P_{2,\mathfrak{p}} \end{array}$$

As P is a finitely generated projective module, its localization at a prime ideal is free, hence each $P_{i,\mathfrak{p}}$ has a basis. The key is that the projection onto $P_{1,\mathfrak{p}}/I_{\mathfrak{p}}P_{1,\mathfrak{p}}$ is still a basis. If we show then as $\varphi_{\mathfrak{p}}$ is an isomorphism, by commutativity the image of the basis in $\tilde{\varphi}_{\mathfrak{p}}$ must be linearly independent which shows the map is injective. To that end, if $\{e_1, \dots, e_n\}$ is a basis for $P_{1,\mathfrak{p}}$, its projection spans so let's show it is linearly independent. Say

$$\sum_i r_i \bar{e}_i = 0$$

Then $\sum_i r_i e_i \in I_{\mathfrak{p}}P_{1,\mathfrak{p}}$. By properties of ideals, $r_i \in I_{\mathfrak{p}}$, but then we must have that in the quotient $r_i = 0$ for all i , which by our previous reasoning allows us to conclude injectivity.

Hence, $\tilde{\varphi}_{\mathfrak{p}}$ is injective for each prime $\mathfrak{p} \subseteq R$, and so it is a classical algebra result that $\tilde{\varphi}$ is injective (see [2, chapter 20.3]). $\square$

Now, let us show that if I is a nilpotent ideal, then there is a bijection between the finitely generated projective R -modules and the finitely generated projective R/I -modules, which shall then complete the proof. Without loss of generality we can assume that $I^2 = 0$ (we can then use induction to for the result for $I^n = 0$ for $n \geq 2$)^a. Let us first show that an idempotent $\bar{e} \in R/I$ lifts to an idempotent $e \in R$. If r is a lift of $\bar{e}$, then we can verify that:

$$e = r + rxr + (1 - r)y(1 - r)$$

where $x, y \in I$. Verify that $e^2 = e$ and it indeed maps to $\bar{e}$. This lift is unique up to conjugation by an element $u \in 1 + I$. With this lift, we shall show that every finitely generated R/I -module $\bar{P}$ is isomorphic to P/IP for an appropriate finitely generated projective module P . The key is that idempotent matrices correspond to summands of free modules. In particular if $\bar{P}$ is a finitely generated projective R/I -module, then there exists an idempotent $\bar{e} \in M_n(R/I)$ such that $\text{im}(\bar{e}) = \bar{P}$. Then this $\bar{e}$ can be lifted to $e \in M_n(R)$ where $e^2 = e$ and $e \equiv \bar{e} \pmod{I}$, so by taking $\text{im}(e) = P$ we get:

$$\bar{P} \cong P/IP$$

Finally, we are at the stage where we reduced our problem of finitely generated projective R -modules to modules of the form P/IP . Then by our lemma, $P_1/IP_1 \cong P_2/IP_2$ if and only if $P_1 \cong P_2$, showing that the isomorphism classes are in bijective corresponds, completing the proof.

^aThis is where the completeness generalization also comes in, as we can consider $\varprojlim R/I^n$

Example 2.3: 0-dimensional Commutative Ring

Let R be a commutative ring and R_{red} be its reduction. Then it is an easy exercise to show that R_{red} is Artinian if and only if $\text{Spec}(R)$ is finite and discrete. More generally (in the “non-finite case”), if a ring is von Neumann regular (for each $x \in R$, there exists a $y \in R$ such that $xyx = x$; this is a “weak inverse”), then R has Krull dimension 0 and is reduced (in fact, this is an equivalent characterization of von-Neumann regular rings, giving a good geometric characterization of such rings^a)

^aNote that there are many non-commutative examples, such matrix rings, left principal ideals generated by idempotents, and $\text{End}_k(V)$ for a vector space, and as they are closed under direct product that makes every semi-simple von-Neumann regular ring semi-simple

2.3 Connection to H_0

We start connecting the K_0 ring to (co)homological results, starting with the most simple one of the $H_0(R)$ group, which we shall take as $\text{Hom}(\text{Spec}(R), \mathbb{Z})$. Recall that for a ring R , there is a natural map $\text{Spec}(R) \rightarrow \mathbb{Z}$ sending $\mathfrak{p} \mapsto \dim_{\kappa(\mathfrak{p})} M \otimes_R \kappa(\mathfrak{p})$. To see this connection, consider the following warm-up result:

Theorem 2.3.1: Pierce’s Theorem

Let R be a 0-dimensional commutative ring. Then:

$$K_0(R) = \text{Hom}(\text{Spec}(R), \mathbb{Z})$$

Note that no assumption was made on R be finitely generated, for example it may be an infinite product of fields.

Proof :

Recall that as R is 0-dimensional all its prime ideals are maximal, and $\text{Spec}(R)$ has the discrete topology. Then by generalizing the classification of Artinian rings to 0-dimensional rings (in particular, every finitely generated projective R -module will be a finite product of modules of the form $R/\mathfrak{m}$), we can consider the rank of each $R/\mathfrak{m}$ individually which gives us the exact information of the $K_0(R)$ group, completing the proof.

We shall thus push this one step further and see the relation for more general commutative rings:

Definition 2.3.1: H_0 Group

Let R be a commutative ring. Then:

$$H_0(R) = [\text{Spec}(R), \mathbb{Z}] = \text{Hom}(\text{Spec}(R), \mathbb{Z})$$

As $\text{Spec}(R)$ is (quasi)compact, there are at most finitely many components, and so $H_0(R)$ will always be a free abelian group (ex. it can’t be $\mathbb{Q}$). If furthermore R is Noetherian, then:

$$H_0(R) \cong \mathbb{Z}^n$$

If on top of that R is an (integral) domain (more generally $\text{Spec}(R)$ is connected), then:

$$H_0(R) \cong \mathbb{Z}$$

Lemma 2.3.1: H_0 direct summand of K -group

The group $H_0(R)$ is a direct summand of $K_0(R)$. If $\dim R = 0$, then $K_0(R) \cong H_0(R)$.

Proof :

Take the monoid L of the componentwise free modules R^n . Then as we saw, L is cofinal, and $L \rightarrow P(R)$ is a semiring map which naturally gives a map $L^{-1}L \rightarrow K_0(R)$. Then as L is isomorphic to $\text{Hom}(\text{Spec}(R), \mathbb{N})$, we must have $L^{-1}L \cong H_0(R)$, and it a subgroup.

Now, there is a natural map $K_0(R) \rightarrow H_0(R)$ given by the rank, namely there is a natural map from $\text{rank} : P(R) \rightarrow [\text{Spec}(R), \mathbb{N}]$, which is a semiring homomorphism. Thus, we get $H_0(R) \rightarrow K_0(R) \rightarrow H_0(R)$, is the identity, and so $H_0(R)$ is a direct summand of $K_0(R)$, as we sought to show.

For the special case of $\dim(R) = 0$, By Pierce's Theorem we have $K_0(R) \cong H_0(R)$.

As $K_0(R)$ splits with $H_0(R)$, the other component is the kernel of the rank map. We shall denote it $\tilde{K}_0(R)$ so that:

$$K_0(R) \cong H_0(R) \oplus \tilde{K}_0(R)$$

It shall be shown later that $\tilde{K}_0(R)$ is a nilradical, and as $H_0(R)$ is reduced already we would have that it is the nilradical of $K_0(R)$.

Lemma 2.3.2: Limit and Kernel

Let R be a commutative, let $P_n(R)$ denote the subset of all modules in $P(R)$ consisting of projective modules of constant rank n . There is a map $P_n(R) \rightarrow K_0(R)$ sending

$$P \mapsto [P] - [R^n]$$

This map is compatible with the stabilization map $P_n(R) \rightarrow P_{n+1}(R)$ sending $P \mapsto P \oplus R$, and the induced map is an isomorphism:

$$\varinjlim_n P_n(R) \cong \tilde{K}_0(R)$$

Proof :

Follows from 2.1.1.

Though we could take the colimit, by the base cancellation theorem we know that the interesting “geometric” structure of a finitely generated projective R -module over a ring of rank d can only happen for modules of rank $\leq d$:

Corollary 2.3.1: Higher Dimension Nilradical

Let R be a commutative Noetherian ring where $\text{kdim } R = d$. Then for every $n > d$,

$$P_n(R) \cong \tilde{K}_0(R)$$

Proof :

Let P, Q be finitely generated projective modules of rank $> d$. Then by the Base cancellation Theorem (theorem 1.1.1), we get that:

$$[P] = [Q] \text{ if and only if } P \cong Q$$

giving the desired result.

The H_0 group is based off of the spectrum of R , and hence is dependent on its commutativity. Let us now try to bring this interpretation to non-commutative rings. To do this, another interpretation of the kernel that is rather nice is that it is the intersection of all the kernels of map $K_0(R) \rightarrow K_0(F)$ for all maps $R \rightarrow F$ into a field: as rank is a natural map, $\tilde{K}_0(F) = 0$, and $H_0(R)$ captures the information at each prime ideals (i.e. each stalk), where we can get this information by going over all fields for which we can have $R \rightarrow F$.

We may thus take the non-commutative equivalent of a field: a simple Artinian ring. As maximal commutative subrings of a simple Artinian ring S is a finite product of 0-dimensional local rings, we get that this is the correct generalization. We can define $\tilde{K}_0(R)$ as the intersection of the kernels of $K_0(R) \rightarrow K_0(S)$ where S is a simple Artinian ring. If no map $R \rightarrow S$ exists, then we say $\tilde{K}_0(R) = K_0(R)$. We define:

$$H_0(R) = \frac{K_0(R)}{\tilde{K}_0(R)}$$

We saw that $H_0(R)$ is a free-abelian group in the commutative case. In the non-commutative case it is at least still torsion-free. To see this, let X be the set of homomorphisms $R \rightarrow S_x$ through which any other $R \rightarrow S'$ factors (such a set exists by using Zorn's lemma). Then as $K_0(S_x) \rightarrow K_0(S')$ is an isomorphism. Hence, $\tilde{K}_0(R)$ is the intersection of all kernels of maps $K_0(R) \rightarrow K_0(S_x)$. We also then have that $H_0(R)$ is the image of $K_0(R)$ in the torsion free group:

$$\prod_{x \in X} K_0(S_x) \cong \prod_{x \in X} \mathbb{Z}$$

(The author of the textbook I'm reading talks a bit about whitehead groups, which I am omitting for now)

2.3.1 Some Important Functionals

We are in the realm of representation theory, hence we have our usual trace and determinant map that shows up. In this section, we shall see how these maps give us some interesting information on our K_0 and H_0 group.

Hattori-Stallings Trace Map

Let R be an associative ring, and let $[R, R] = \{rs - sr \mid r, s \in R\}$ be the commutator subgroup. Then the trace provides a natural map:

$$\text{tr} : M_n(R) \rightarrow \frac{R}{[R, R]}$$

where we quotient by $[R, R]$ as tr should be independent of conjugation, which shall require $a_{i\ell}b_{\ell i}$ to commute. Then it should be a simple computation to see that all traceless matrices are in

$[M_n(R), M_n(R)]$ which gives us the isomorphism:

$$\frac{M_n(R)}{[M_n(R), M_n(R)]} \cong \frac{R}{[R, R]}$$

Naturally, not all elements of $\text{End}_R(P)$ of a finitely generated projective R -module P is in matrix form, however it close enough to still define the trace: as P is finitely generated and projective choose Q such that $P \oplus Q \cong R^n$. Then we have an idempotent element $e \in M_n(R)$ such that $e(R^n) = P$, so:

$$\text{End}_R(P) = eM_n(R)e$$

It is not too hard to verify that any other choices of idempotent will yield an e_1 which is idempotent and conjugate to e in some $M_m(R)$. Thus, we have that the trace map is independent of representative and we have a well-defined map:

$$\text{tr} : \text{End}_R(P) \rightarrow \frac{R}{[R, R]}$$

The trace of the identity map on P will be the trace of the idempotent element e ; we call this particular value the *trace* of P .

Now, if P' is a finitely generated projective R -module represented by an idempotent f , then $P \oplus P'$ is represented by $\begin{pmatrix} e & 0 \\ 0 & f \end{pmatrix}$, and so we have its trace is $\text{tr}(P) + \text{tr}(P')$! Thus, the trace is an additive map on $P(R)$, and so we get a natural map:

Definition 2.3.2: Hattori-Stallings Trace Map

Let R be a (not necessarily commutative ring). Then by the above discussion, the natural map:

$$K_0(R) \rightarrow \frac{R}{[R, R]}$$

is called the *Hattori-Stallings Trace Map*.

In the case where R is commutative, we have that $R = \frac{R}{[R, R]}$. This gives a more explicit description of $K_0(R)$ and $H_0(R)$ (as we have $\text{Spec}(R)$ well-defined). To see this, start by taking the trace map on $K_0(R) \rightarrow R$ and restrict to $H_0(R)$. Then notice that each $\text{Spec}(R) \rightarrow \mathbb{Z}$ gives a decomposition of $\text{Spec}(R)$ into components $R = R_1 \times \cdots \times R_c$ given by idempotents $e_1, \dots, e_c$. As $\text{tr}(e_i R) = e_i$, we have that for any $f \in R$,

$$\text{tr}(f) = \sum_i f(i)e_i$$

And so we get:

$$\text{tr}(fg) = \text{tr}(f) \text{tr}(g)$$

We thus get the following natural map:

Proposition 2.3.1: Hattori-stallings Map Factorization

Let R be *commutative*. Then the Hattori-Stallings trace map factors through:

$$K_0(R) \xrightarrow{\text{rank}} H_0(R) \rightarrow R$$

Proof :

Consider the following diagram:

$$\begin{array}{ccc} K_0(R) & \longrightarrow & \prod_{\mathfrak{p}} K_0(R_{\mathfrak{p}}) \\ \text{tr} \downarrow & & \downarrow \text{tr} \\ R & \xhookrightarrow{\text{diag}} & \prod_{\mathfrak{p}} R_{\mathfrak{p}} \end{array}$$

As the kernel of the top arrow is $\tilde{K}_0(R)$ by construction, we get the left arrows map to $H_0(R)$, which gives the desired factorization.

Example 2.4: Trace and Characters on Representations

1. Let k be commutative and $R = k[G]$ of a group G . Then if $g, h \in G$ are conjugate, then $g - h \in [R, R]$ (as $g = xhx^{-1}$ so $xhx^{-1} - h \in [xh, x^{-1}]$). Then we see that $\frac{R}{[R, R]} \cong \oplus k[g]$ for $g \in G/\sim$ (i.e. up to conjugacy class). Given such a basis, we then get:

$$\text{tr}(P) = \sum_{g \in G} r_P(g)[g]$$

where $r_P(g)$ are coefficients seen as functions of the set $G/\sim$, for each P .

2. In the special case where G is finite, then any finitely generated projective $k[G]$ -module P is also a projective k -module, and so we can form the trace map $\text{End}_k(P) \rightarrow k$! Hence, we get the character $\chi_P : G \rightarrow k$ given by the formula:

$$\chi_P(g) = \text{tr}(g)$$

It was proven by a mathematician called *Hattori* that we can write:

$$\chi_P(g) = |Z_G(g)|r_P(g^{-1})$$

which gives a very interesting connection between characters and the general theory of trace we've been building up! We shall not look into this more deeply, but it is an interesting result to point out.

One very interesting result we get out of this is that $\mathbb{Z}[G]$ does not split, i.e. it has no non-trivial idempotents:

Proof. Let $e \in \mathbb{Z}[G]$ be an idempotent element. Then $\chi_P(1)$ is the rank of the $\mathbb{Z}$ -module $P = e\mathbb{Z}[G]$. Then by construction its rank is $\leq |G| = \text{rank } \mathbb{Z}[G]$. As $r_P(1) \in \mathbb{Z}$, this contradicts Hattori's formula as $\chi_P(1) = |G|r_P(1)$, completing the proof. $\square$

3. If k is a field of characteristic 0 and G a finite group, then by Maschke's Theorem we have $k[G]$ is the simple product of k -algebras $S_1 \times \cdots \times S_c$ where c represents the conjugacy classes of G . Hence we get:

$$\frac{k[G]}{[k[G], k[G]]}$$

Then we saw in example 2.2 that

$$K_0(K[G]) \cong \mathbb{Z}^c$$

By Hattori's formula, we can see that the trace map from $K_0(K[G])$ is isomorphic to $\mathbb{Z}^c \subseteq k^c$.

Determinant

Recall definition 1.2.3 where we can define the determinant in a natural way if R is commutative on a finitely generated projective R -module. We saw in proposition 1.2.4 that we may classify all finitely generated projective R -modules over a 1-dimension ring, and in proposition 1.2.4 we saw that the determinant of a finitely generated projective R -module P can be seen as elements of $\text{Pic}(R)$. These results work when reducing to K -groups:

Proposition 2.3.2: K_0 Group and Determinant

Let R be a commutative ring. Then the determinant determines a natural map:

$$\det : K_0(R) \rightarrow \text{Pic}(R)$$

Proof :

The key is that the non-top wedge products disappear. By the universal properties of monoids, it suffices to show that:

$$\det(P \oplus Q) \cong \det(P) \otimes \det(Q)$$

Without loss of generality, we may assume that P, Q have constant rank n, m respectively. Then $\bigwedge^{n+m} P \oplus Q$ is the sum over all i, j such that $i + j = n + m$ in $(\bigwedge^i P) \otimes (\bigwedge^j Q)$. Then if $i > n$ and $j > m$ then either $\bigwedge^i P = 0$ or $\bigwedge^j Q = 0$. But then:

$$\bigwedge^{n+m} P \oplus Q \cong \left(\bigwedge^n P \right) \otimes \left(\bigwedge^m Q \right)$$

as we sought to show.

Corollary 2.3.2: Rank and Determinant Represents K -group

Let R be a commutative ring. Then $H_0(R) \oplus \text{Pic}(R)$ is a ring with a square-zero ideal $\text{Pic}(R)$ and there is a surjective homomorphism:

$$K_0(R) \twoheadrightarrow H_0(R) \oplus \text{Pic}(R)$$

with kernel consists of equivalence classes representable by $[P] - [R^n]$ where P has constant rank n and $\bigwedge^n P \cong R$.

Proof :

Let $SK_0(R)$ represent the kernel. Notice first that this is naturally the kernel of $\det : K_0(R) \rightarrow \text{Pic}(R)$ by the above proposition. Next, $SK_0(R)$ is an ideal of $K_0(R)$. After showing this, the rest is combining earlier results.

Corollary 2.3.3: Rank and Determinant: One Dimensional Case

Let R be a commutative Noetherian 1-dimensional ring. Then from the above we have:

$$K_0(R) \cong H_0(R) \oplus \text{Pic}(R)$$

Proof :

This follows from the above corollary and proposition 1.2.3.

2.4 Morita Equivalence

An important invariant in the study of modules is the one given by the *Morita Equivalence*, which tells you when a ring action produces the same theory:

Definition 2.4.1: Morita Equivalence

Let R, S be (not necessarily commutative) rings. Then $R\text{-}\mathbf{Mod}$ and $S\text{-}\mathbf{Mod}$ are *Morita equivalent* if they are equivalent, namely there exists functors F, G such that $GF \cong \text{id}_R$ and $FG \cong \text{id}_S$ where id_R and id_S are the identity functors on their respective categories

We have in fact already seen many examples of Morita equivalence without invoking it, as we shall see in examples 2.5.

Note that F, G being inverses implies they preserve limits and colimits, hence the structure of in equivalence categories share many similarities (for example, direct sums and products pass through F, G).

We can in fact have a precise idea of what is the image of F, G . Let $F(R) = P$ and $G(S) = Q$. Then notice we have the natural maps:

$$R \cong \text{End}_R(R)^{\text{op}} \xrightarrow{F} \text{End}_S(P)^{\text{op}} \quad \text{and} \quad S \cong \text{End}_S(S)^{\text{op}} \xrightarrow{G} \text{End}_R(Q)^{\text{op}} \quad (2.1)$$

which make P into an S, R -bimodule and Q into a R, S -bimodule (if this is unfamiliar to you, see [2, chapter 23.2] for a review of the relation between R and its endomorphisms). Now, since F, G preserve limits/colimits, $F(R^\oplus) = P^\oplus$ and $G(S^\oplus) = Q^\oplus$, and any relation is preserved as equalizers are preserved, hence we have that for any $M \in \text{Obj}(R\text{-}\mathbf{Mod}), N \in \text{Obj}(S\text{-}\mathbf{Mod})$

$$F(M) \cong P \otimes_R M \quad \text{and} \quad G(N) \cong Q \otimes_S N$$

as bimodules, and:

$$GF(R) = G(P) = Q \otimes_S P \stackrel{!}{\cong} R \quad \text{and} \quad FG(S) = F(Q) = P \otimes_R Q \stackrel{!}{\cong} S$$

giving a very structured result for Morita equivalent categories! The $\stackrel{!}{\cong}$ comes from how we can think of the endomorphisms as being the “negative dimensional” lines bundles, or for more rigor you can checkout [2, chapter 24.2]. The following summarizes our above discussion and adds some useful insights the user may appreciate:

Theorem 2.4.1: Structure Theorem For Morita Equivalence

Let R, S be Morita Equivalent rings (represented by the functors F, G) and $F(R) = P$, $G(S) = Q$. Then:

1. P, Q are finitely generated projective both as R -modules and S -modules
2. $\text{End}_S(P) \cong R \cong \text{End}_S(Q)^{\text{op}}$ and $\text{End}_R(Q) \cong S \cong \text{End}_R(P)^{\text{op}}$
3. P, Q are dual to each other as S -modules:

$$P \cong \text{Hom}_S(Q, S) \quad \text{and} \quad Q \cong \text{Hom}_S(P, S)$$

The tensor gives us the R -tensor equivalent. Furthermore, $\text{Hom}_S(P, -)$ and $\text{Hom}_R(Q, -)$ (note it is over R) are faithful functors from **Mod** to **Ab** (for the respective rings).

4. The image of F, G are isomorphic (as bimodules) to:

$$F(M) \cong P \otimes_R M \quad \text{and} \quad G(N) \cong Q \otimes_S N$$

Proof :

1. This comes from $Q \otimes_S P \cong_R R$ and $P \otimes_R Q \cong_S S$
2. Follows from equation 2.1.
3. summarizes the above discussion
4. ibid

Corollary 2.4.1: Morita Equivalence and K Group

Let R, S be Morita Equivalent. Then

$$K_0(R) \cong K_0(S)$$

Proof :

Since P, Q are finitely generated and projective, and F, G represent the equivalence, we get

$$P(R) \cong P(S)$$

giving the desired result

Naturally, having isomorphic K -groups does not imply Morita equivalence (think $K_0(\mathbb{Z})$ and $K_0(\mathbb{Q})$).

Example 2.5: Morita Equivalence

The prototypical example is R and $M_n(R)$ for $n \geq 1$. In this case, $F(R) = P$ is the bimodule R^n of “column vectors” while $Q(M_n(R)) = (R^n)^t$ is the bimodule of “row vectors”.

More generally, by theorem 2.4.1(3), if P is a faithful finitely generated projective R -module,

then $\text{End}_R(P)$ is Morita Equivalent to R .

2.5 *Mayer-Vietoris Sequence

In algebraic topology, the Mayer-Vietoris sequence is a useful sequence for computing homotopic groups by gluing together pieces. We bring over this discussion for computing K groups when gluing user Milnor squares. As we saw via the Hattori-Stallings Trace Map allows for some very important decompositions:

$$R \rightarrow \frac{R}{[R, R]}$$

As we just saw, $M_n(R)$ produces equivalent categories. More generally, in representation theory we are interested in $\text{GL}_n(R)$, the general linear group. Which n should we pick? Can we relate the n 's in Chapter 3, we shall take the limit over all the n to define the group $\text{GL}(R)$. For now, we relate these through the natural map:

$$g \mapsto \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix}$$

These groups shall be central in understanding K_1 in the next section. One important construction we shall use is patching together information using Milnor patching: consider the Milnor square:

$$\begin{array}{ccc} R & \xrightarrow{f} & S \\ \downarrow & & \downarrow \\ R/I & \xrightarrow{\bar{f}} & S/I \end{array}$$

Recall theorem 1.1.2(3) that any finitely generated projective R -module can derive from patching, hence define

$$\partial_n : \text{GL}_n(S/I) \rightarrow K_0(R)$$

where $\partial_n(g) = [P] - [R^n]$. By theorem 1.1.2(4), we see that:

$$\partial_n(g) = \partial_{n+1} \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix}$$

Hence, taking the sequence of $\{\partial_n\}_{n=1}^\infty$ give a homomorphism:

$$\partial : \text{GL}(S/I) \rightarrow K_0(R)$$

Then from Milnor-Patching

Theorem 2.5.1: Mayer-Vietoris

Given a Milnor square as above, the sequence

$$\text{GL}(S/I) \xrightarrow{\partial} K_0(R) \xrightarrow{\Delta} K_0(S) \oplus K_0(R/I) \xrightarrow{\pm} K_0(S/I)$$

is exact. The image of ∂ is the double coset space

$$\text{GL}(S) \backslash \text{GL}(S/I) / \text{GL}(R/I) = \text{GL}(S/I) / \sim$$

where $x \sim gxh$ for $x \in \text{GL}(S/I)$, $g \in \text{GL}(S)$, and $h \in \text{GL}(R/I)$.

Proof :

Follows from theorem 1.1.2(4) and our above discussion.

The Mayer-Vietoris sequence is a useful computation tool, so let's show how it can be used:

Example 2.6: Mayer-Vietoris

Consider the coordinate ring of an irreducible curve with a node (which can be shown is unique up to projective equivalence^a). Let us compute K_0 . As this is a 1-dimensional Noetherian ring, by corollary 2.3.3 we have:

$$K_0(R) \cong H_0(R) \oplus \text{Pic}(R)$$

Then the result follows from the following exact sequence known as the *Units-Pic sequence*:

Let Δ denotes the diagonal map, $\pm$ the difference map sending $(s, \bar{r})$ to $\bar{s}f(\bar{r})^{-1}$, resp. (L', L) to $L' \otimes_S S/I \otimes_{R/I} L^{-1}$. Then given a Milnor square, the following is exact:

$$1 \rightarrow R^\times \xrightarrow{\Delta} S^\times \times (R/I)^\times \xrightarrow{\pm} (S/I)^\times \xrightarrow{\partial}$$

$$\text{Pic}(R) \xrightarrow{\Delta} \text{Pic}(S) \times \text{Pic}(R/I) \xrightarrow{\pm} \text{Pic}(S/I).$$

Proof. As R is the pullback of S and R/I , exactness at the first two places is immediate. Theorem 1.1.2 immediately gives exactness at the last two places, leaving only exactness at $(S/I)^\times$. Given $s \in S^\times$ and $\bar{r} \in (R/I)^\times$, set $\beta = \pm(s, \bar{r}) = \bar{s}f(\bar{r})^{-1}$, where $\bar{s}$ denotes the reduction of s modulo I . Then we see that $\lambda = (s, \bar{r}) \in L_\beta \subset S \times R/I$, and every element of L_β is a multiple of λ . It follows that $L_\beta \cong R$. Conversely, suppose given $\beta \in (S/I)^\times$ with $L_\beta \cong R$. If $\lambda = (s, \bar{r})$ is a generator of L_β , then I claim that s and $\bar{r}$ are units, which implies that $\beta = \bar{s}f(\bar{r})^{-1}$ which completes the proof. To that end, if $s' \in S$ maps to $\beta \in S/I$, then $(s', 1) \in L_\beta$, namely since $(s', 1) = (xs, x\bar{r})$ for some $x \in R$, this implies that $\bar{r} \in (R/I)^\times$. If $t \in S$ maps to $f(\bar{r})^{-1}\beta^{-1} \in S/I$, then $st \equiv 1$ modulo I . But then $I \subset sR$ because $I \times 0 \subset L_\beta$, so $st = 1 + sx$ for some $x \in R$. So $s(t - x) = 1$, and hence $s \in S^\times$ giving the desired result. $\square$

Hence, we get that:

$$K_0(R) \cong \mathbb{Z} \oplus k^\times$$

^aEquivalence after a linear isomorphism

2.6 Topological K-Group

Let us now take a moment to examine the K-theory on the spaces explored in section 1.3. We shall build-up to an interesting result known as *Bot Periodicity*. First, we can define the homotopy groups of U and O by:

$$\pi_k(U) = \pi_k(\varinjlim_n U(n)) \quad \pi_k(O) = \pi_k(\varinjlim_n O(n))$$

which is well-defined as the fibrations

$$U(n-1) \hookrightarrow U(n) \rightarrow S^{2n-1} \quad O(n-1) \hookrightarrow O(n) \rightarrow S^{n-1}$$

naturally induces the inclusion $\pi_k(U(n-1)) \rightarrow \pi_k(U(n))$ and $\pi_k(O(n-1)) \rightarrow \pi_k(O(n))$. Looking at these groups, we get the following fascinating result:

Theorem 2.6.1: Bott Periodicity

Let U, O be above groups. Then the homotopy groups of U, O are:

$k \pmod{8}$	0	1	2	3	4	5	6	7
$\pi_k(U)$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$
$\pi_k(O)$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	0	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$

In particular, we see that $\pi_k(U)$ is 2-periodic, and $\pi_k(O)$ is 8-periodic.

Proof :

Delayed till ref:HERE

The details of this proof is beyond the scope for this book, but an outline of the idea shall be presented.

2.6.1 K-groups over different fields

From section 1.3 recall the collection of isomorphism classes of vector bundles on a nonempty¹ paracompact space X is denoted $\mathbb{V}\mathbb{B}_{\mathbb{R}}(X)$, $\mathbb{V}\mathbb{B}_{\mathbb{C}}(X)$, or $\mathbb{V}\mathbb{B}_{\mathbb{H}}(X)$ depending on whether it is for real, complex, or quaternion vector bundles. We shall focus on the real and complex case. Certainly under Whitney sums the real and complex vector bundles form an abelian monoid. As we can define the tensor product (the point-wise definition extends naturally, see the beginning of section 1.3 or [5]). Hence, we can define the Grothendieck completions of $\mathbb{V}\mathbb{B}_{\mathbb{R}}(X)$ and $\mathbb{V}\mathbb{B}_{\mathbb{C}}(X)$ and get:

$$KO(X) \quad KU(X)$$

give commutative rings with identity being the trivial line bundles. If it is clear over which field we are, we may write $K(X)$. We have omitted $KSp(X)$ as $\mathbb{H}^m \otimes_{\mathbb{H}} \mathbb{H}^n \cong \mathbb{H}^{mn}$. However,

$$\mathbb{H}^m \otimes_{\mathbb{R}} \mathbb{R}^n \cong \mathbb{H}^{mn}$$

and so it has a natural $\mathbb{R}$ -module structure.

Some immediate results we can get from algebraic topology

1. if $X = \{*\}$ is the one point space (or more generally a contractible space as vector bundle isomorphism classes are defined up to homotopy), then we are working with the usual K -theory and so:

$$KO(*) = KU(*) = \mathbb{Z}$$

2. If $f : X \rightarrow Y$ is continuous, the induced line bundle map $E \mapsto f^*(E)$ gives a morphism of monoids and semi-rings

$$f^* : \mathbf{VB}(Y) \rightarrow \mathbf{VB}(X)$$

which then induces

$$f^* : K(Y) \rightarrow K(X)$$

Thus, as there is a universal map $f : X \rightarrow *$, there is a universal map from $K(*) = \mathbb{Z} \rightarrow K(X)$. As we are assuming $X \neq \emptyset$, we may also define a map $*$ $\rightarrow X$ which can be compose with

¹If it is empty, the vector bundle is always trivial, and it will be a bit of a headache to mention it constantly

the above universal map and which splits, and hence we further get that $\mathbb{Z}$ is in fact a direct summand of $K(X)$

3. The direct sum of trivial bundles will form a cofinal object when X is compact (namely since any map $X \rightarrow \mathbb{N}$ can only take finitely many values as the has to be finitely many components, and the subbundle of a bundle always has an orthogonal compliment and so can be decomposed into 1-dimensional trivial bundles, but then $[X, \mathbb{N}]$ and $\mathbb{N}$ would both be cofinal). From this, we can just like in section 2.3 have the subrings:

$$\mathbb{Z} \subseteq [X, \mathbb{Z}] \subseteq K(X)$$

and the “splitting” map:

$$\dim : K(X) \rightarrow [X, \mathbb{Z}]$$

which gives us:

$$K(X) \cong \tilde{K}(X) \oplus [X, \mathbb{Z}]$$

where $\tilde{K}(X)$ is the kernel of the dimension map.

Let us now take a look at the classifying spaces BO and BU and their K_0 -group.

Theorem 2.6.2: Stabilization Theorem of Classifying Spaces

Let X be either a compact space or a finite-dimensional connected CW complex. Then the map (3.1.0) induces an isomorphism $\tilde{K}(X) \cong \varinjlim \mathbf{VB}(X) \cong \varinjlim [X, \text{Grass}_n]$. In particular,

$$\widetilde{KO}(X) \cong \varinjlim [X, BO_n], \quad \widetilde{KU}(X) \cong \varinjlim [X, BU_n],$$

and

$$\widetilde{KSp}(X) \cong \varinjlim [X, BSp_n].$$

Proof :

Since the monoid of (componentwise) trivial vector bundles is cofinal in $\mathbf{VB}(X)$, we shall show that every element of $\tilde{K}(X)$ is represented by an element

$$[E] - n$$

of some $\mathbf{VB}_n(X)$ by lemma 2.1.1. Now, if $[E] - n = [F] - n$, then $E \oplus T^n \cong F \oplus T^n$ in some $\mathbf{VB}_{n+\ell}(X)$. But that E is stably free for large enough n , and so we must have

$$\tilde{K}(X) \cong \varinjlim \mathbf{VB}_n(X)$$

As this is the case for each O, U, Sp , we get our desired result.

2.6.2 Bott Periodicity

Let us return to the setting of paracompact spaces X with structure groups O_n , U_n , or Sp_n respectively. The vector bundles are given by the classifying spaces BO , BU and BSp which can be realised as

the natural vector bundle on the limit of Grassmannians $\text{Grass}_n(V)$ for finite dimensional V of $\mathbb{R}^\infty$, $\mathbb{C}^\infty$ or $\mathbb{H}^\infty$ (depending on the type of vector bundle). Then Bott periodicity can be stated as:

$$BU \times \mathbb{Z} \cong \Omega^2(BU \times \mathbb{Z}) \quad BO \times \mathbb{Z} \cong \Omega^8(BO \times \mathbb{Z})$$

where Ω^n is the n -dimensional loop space (i.e. the space not up to homotopy). In Bott's original proof, he uses spaces known as "symmetric spaces" which are compact Riemann manifolds such that for every point $p \in M$, there is an isometry ι_p that fixes p and reverses the geodesics through p (hence a symmetric space). The sphere gives rise to such a space, and by taking homotopies we get U, O, Sp . Here, Bott looks at a new spaces M^ν where $\nu = (P, Q, h)$ is a triple of two points in M and h a homotopy class of curves connecting P, Q , and M^ν is the set of all geodesics of minimal length which join P and Q that are in h . Then he showed that:

$$\pi_k(M^\nu) = \pi_{k+1}(M)$$

For appropriate dimension considerations (see Bott's paper for more details). From this, he managed to conclude:

$$\pi_k(U) \cong \pi_{k+1}(BU) \cong \pi_{k+2}(U)$$

and by manipulating U , O , and Sp , we also get:

$$\pi_k(O) \cong \pi_{k+1}(BO) \cong \pi_{k+2}(U/O) \cong \pi_{k+3}(Sp/U) \cong \pi_{k+4}(Sp)$$

$$\pi_{k+5}(BSp) \cong \pi_{k+6}(U/Sp) \cong \pi_{k+7}(O/U) \cong \pi_{k+8}(O)$$

which would give our desired periodicity! The details are beyond this book, but it should be enlightening to see how this theory results are rather simple to calculate consider how in general it is very difficult to calculate homotopy groups, for example here are the known homotopy groups for spheres:

	π_1	π_2	π_3	π_4	π_5	π_6	π_7	π_8	π_9	π_{10}	π_{11}	π_{12}	π_{13}	π_{14}	π_{15}
S^1	$\mathbb{Z}$	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S^2	0	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{12}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_{15}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_{12} \times \mathbb{Z}_2$	$\mathbb{Z}_{84} \times \mathbb{Z}_2^2$	$\mathbb{Z}_2^2$
S^3	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{12}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_{15}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_{12} \times \mathbb{Z}_2$	$\mathbb{Z}_{84} \times \mathbb{Z}_2^2$	$\mathbb{Z}_2^2$
S^4	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z} \times \mathbb{Z}_{12}$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_{24} \times \mathbb{Z}_3$	$\mathbb{Z}_{15}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_{120} \times \mathbb{Z}_{12} \times \mathbb{Z}_2$	$\mathbb{Z}_{84} \times \mathbb{Z}_2^5$
S^5	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{30}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_{72} \times \mathbb{Z}_2$
S^6	0	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_{60}$	$\mathbb{Z}_{24} \times \mathbb{Z}_2$	$\mathbb{Z}_2^3$
S^7	0	0	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	0	0	$\mathbb{Z}_2$	$\mathbb{Z}_{120}$	$\mathbb{Z}_2^3$
S^8	0	0	0	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	0	0	$\mathbb{Z}_2$	$\mathbb{Z} \times \mathbb{Z}_{120}$

Figure 2.1: Known Homotopy Groups; yellow is tentative (see this link)

In fact, we can see the results of Bott periodicity in this graph with the stability of the results for as n get's large for $\pi_{n+k}(S^n)$ where $k \geq 2$, and how this result it much harder for the rest of the cases!

3

K_1 Group

The K_0 group let us classify the vector bundles over a ring or (paracompact) topological space. We saw in section 2.6 that for (say complex) vector bundles over X :

$$K_0(X) = [X, \mathbb{Z} \times BU]$$

Showing there is a relation between homotopy groups and $K_0(X)$. From the algebraic perspective, we found the isomorphism classes of the projective modules. It is now natural to ask for the automorphism classes. We shall focus on the algebraic perspective, but shall see that it will naturally tie with the topological by taking suspension (recall from [5, chapter 5] that there is a natural map $\pi_k(X) \rightarrow \pi_{k+1}(\Sigma X)$).

3.1 Defining K_1

Let R be a ring (with unit). Take the maps:

$$G_n(R) \rightarrow G_{n+1}(R) \quad g \mapsto \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix}$$

The limit of these maps shall be denoted $GL(R)$. Then:

Definition 3.1.1: K_1 Group

Let R be a ring (not necessarily commutative). Then:

$$K_1(R) = \frac{GL(R)}{[GL(R), GL(R)]}$$

Given a ring homomorphism $R \rightarrow S$, there is a naturally induced map $GL(R) \rightarrow GL(S)$ by mapping the elements of R to the diagonal, and as this respects the abelianization we have a natural

map:

$$K_1(R) \rightarrow K_1(S)$$

And hence, $K_1(-)$ is a functor as well.

Before looking at examples, let us have a better grasp of the commutator. As it turns out, these shall always be generated by elementary matrices.

Definition 3.1.2: Elementary Matrices Group of $\text{GL}(R)$

Let $E_n(R)$ denote the elementary matrices of $\text{GL}_n(R)$. Then $E(R)$ denotes the limit of the stabilization morphisms.

Lemma 3.1.1: Elementary Matrix Group is Perfect

For $n \geq 3$, $E_n(R)$ is a perfect group, that is $E_n(R) = [E_n(R), E_n(R)]$

For $n = 2$, we get $\text{SL}_2(\mathbb{F}_2) \cong S_2$ (symmetric group of order 2).

Proof :

For distinct i, j, k , we get:

$$[e_{ij}(r), e_{kl}(s)] = \begin{cases} 1 & \text{if } j \neq k \text{ and } i \neq \ell, \\ e_{i\ell}(rs) & \text{if } j = k \text{ and } i \neq \ell, \\ e_{kj}(-sr) & \text{if } j \neq k \text{ and } i = \ell. \end{cases}$$

and so:

$$e_{ij}(r) = [e_{ik}(r), e_{kj}(1)]$$

Theorem 3.1.1: Whitehead's Lemma

$$K_1(R) = \frac{\text{GL}(R)}{E(R)}$$

Proof :

The commutator subgroup contains $E(R)$ by lemma 3.1.1. Conversely, every commutator in $\text{GL}_n(R)$ can be expressed as a product in $\text{GL}_{2n}(R)$:

$$[g, h] = \begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix} \begin{pmatrix} h & 0 \\ 0 & h^{-1} \end{pmatrix} \begin{pmatrix} (hg)^{-1} & 0 \\ 0 & hg \end{pmatrix}. \quad (3.1)$$

And certainly each of these terms is in $E_{2n}(R)$, giving the desired equivalence.

Example 3.1: K_1 Groups

1. Let R be commutative. Then we have a commutative group homomorphism:

$$\det : \text{GL}(R) \rightarrow R^\times$$

This induces a surjective map $\det : K_1(R) \rightarrow R^\times$ which, for historical reasons, the kernel

is written as $SK_1(R)$. Now, we also have a natural inclusion $R^\times \rightarrow \text{GL}(R)$ mapping to the diagonals. We see that this map is a section map, i.e. it split by the determinant homomorphism. Taking $\text{SL}(R) = \varprojlim_n \text{SL}_n(R)$, as we are working over groups we have the semi-direct product (see [2, chapter 5.3]):

$$\text{GL}(R) \cong \text{SL}(R) \rtimes R^\times$$

Hence, when abelianizing we get:

$$K_1(R) \cong R^\times \oplus SK_1(R)$$

2. If we specifically let $R = k$ be a field where $k \notin \{\mathbb{F}_2, \mathbb{F}_3\}$, then we can show that $\text{SL}(F) = [\text{GL}(F), \text{GL}(F)]$, and so we reduce to:

$$K_1(F) \cong F^\times$$

To see this, we first certainly see that $[\text{GL}(R), \text{GL}(R)] \leq \text{SL}(R)$ as each commutator $[A, B] = ABA^{-1}B^{-1}$ as determinant 1. For the other way around, we must show that each matrix in $\text{SL}_n(R)$ can be written as a product of commutators. As we are working over a field, it suffices to show that all elementary matrices can be written as a product of commutators. But this is now a matter of computations, for example type I elementary matrices $A = I + tE_{ij}$ and $B = I + E_{jk}$ where $k \neq i, j$ (which is possible since $n \geq 3$), then:

$$[A, B] = ABA^{-1}B^{-1} = (I + tE_{ij})(I + E_{jk})(I - tE_{ij})(I - E_{jk}) = I + tE_{ik}$$

notice how it was importance for $n \geq 3$ to insure the computation follow.

3. If $R = R' \times R''$ is a product of rings, then $\text{GL}(R) = \text{GL}(R') \times \text{GL}(R'')$, and the commutator decompose too giving us that products are respected by the K_1
4. The K_1 group is invariant under Morita equivalence. First, as R and $M_n(R)$ are Morita equivalent, we have that:

$$M_{mn}(R) = \text{End}_R(R^m \otimes R^n) \quad M_m(M_n(R)) = \text{End}_{M_n(R)}((M_n(R))^m)$$

And so ,we have a natural isomorphism:

$$\text{GL}_{mn}(R) \cong \text{GL}_m(\text{GL}_n(R))$$

which give the same “stabilization” morphism (the map between $\text{GL}_n(R)$ and $\text{GL}_{n+1}(R)$), and so:

$$K_1(R) \cong K_1(M_n(R))$$

5. Let $R = D$ be a division ring. Then for the same reason as field, every invertible matrix may be reduced to:

$$\text{diag}(r, 1, \dots, 1)$$

and so we have:

$$\frac{\text{GL}_n(R)}{E_n(D)} \cong D^\times / N$$

In lemma 3.1.2, we shall we can reduce to:

$$\frac{\text{GL}_n(R)}{E_n(D)} \cong D^\times / [D^\times, D^\times]$$

for all $n > 1$.

6. Let $F/\mathbb{Q}$ be a finite field extension and $R \subseteq F$ the integrally closed subring (i.e. the algebraic integers). Then it is a result by Bass and Milnor that

$$SK_1(R) = 1$$

and so $K_1(R) = R^\times$. Then by Dirichlet Unit theorem (see [2, chapter 22]), we have that $R^\times$ is isomorphic to:

$$\mu(F) \oplus \mathbb{Z}^{s-1}$$

where $\mu(F)$ is the cyclic group of all roots of unity in F and s is the places at infinity of R , i.e. the number of real embeddings.

Let us take a moment to look at the K_1 group of semi-local rings due to their importance in representation theory (they are the “local rings” in the non-commutative case).

Definition 3.1.3: Semi-local Ring

Let R be a ring. Then R is *semilocal* if $R/J(R)$ is semisimple where $J(R)$ is the (left) Jacobian ideal.

Note that all division rings are semi-local, hence the discussion about semi-local rings generalizes the above discussion.

We shall find a representation of $[R^\times, R^\times]$ for semilocal rings. If $r, s \in R$ are such that $1 + rs$ is a unit, then so is $1 + sr$ because $(1 + sr)(1 - s(1 + rs)^{-1}r) = 1$. Then define subgroup $W(R) \leq R^\times$ generated by the $(1 + rs)(1 + sr)^{-1}$. It is not too hard to show that this subgroup belongs to $E_2(R)$. For $R = M_2(\mathbb{F}_2)$, $W(R) = R^\times \cong \Sigma_3$ and $K_1(R) = 1$ but $R^\times \neq [R^\times, R^\times]$. If T is the subring of upper triangular matrices in $M_2(\mathbb{F}_2)$, its group of units is abelian ($T^\times \cong \mathbb{Z}/2$), but $K_1(T) = 1$ since

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = (1 + rs)(1 + sr)^{-1} \text{ for } r = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \text{ and } s = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Now, a mathematician called *Vaserstein* showed that that $W(R) = [R^\times, R^\times]$ if $\Lambda = R/\text{rad}(R)$ is a product of matrix rings, none of which is $M_2(\mathbb{F}_2)$, and at most one of the factors in Λ is $\mathbb{F}_2$. In particular, $W(R) = [R^\times, R^\times]$ for every local ring R . The same applies for semilocal rings:

Proposition 3.1.1: Environment Title

If R is a semilocal ring, then the natural inclusion of $R^\times = GL_1(R)$ into $GL(R)$ induces an isomorphism $K_1(R) \cong R^\times/W(R)$.

If R is a commutative semilocal ring, then

$$SK_1(R) = 1 \quad \text{and} \quad K_1(R) = R^\times.$$

Proof :

By example 3.1 it suffices to prove that $R^\times$ maps onto $K_1(R)$. This will follow by induction after showing that $GL_n(R) = E_n(R)GL_{n-1}(R)$. Let J denote the Jacobson radical of R , so that R/J is a finite product of matrix algebras over division rings. We will show that $GL_n(R) \twoheadrightarrow GL_n(R/J)$. Indeed, if $I \subseteq R$ is an ideal, each homomorphism $E_n(R) \rightarrow E_n(R/I)$ is onto as the generators $e_{ij}(r)$ of $E_n(R)$ map onto the generators of $E_n(R/I)$ as I is radical. Hence, $(R/J)^\times$ maps onto $K_1(R/J)$; every $\bar{g} \in GL_n(R/J)$ is a product $\bar{e}\bar{g}_1$, where $\bar{e} \in E_n(R/J)$ and $\bar{g}_1 \in GL_1(R/J)$.

Now, given $g \in GL_n(R)$, its reduction $\bar{g}$ in $GL_n(R/J)$ may be decomposed into $\bar{g} = \bar{e}\bar{g}_1$. Then we can lift $\bar{e}$ to an element $e \in E_n(R)$. The matrix $e^{-1}g$ is congruent to the diagonal matrix $\bar{g}_1$ modulo J , so its diagonal entries are all units and its off-diagonal entries lie in J . Using elementary row operations $e_{ij}(r)$ with $r \in J$, it is a matter of computation to reduce $e^{-1}g$ to a diagonal matrix, say to $D = \text{diag}(r_1, \dots, r_n)$. But then the matrix $\text{diag}(1, \dots, 1, r_n, r_n^{-1})$ is in $E_n(R)$. Multiplying D by this matrix yields a matrix in $GL_{n-1}(R)$, giving the induction and the proof.

Lemma 3.1.2: Determinant on semilocal Rings

Let R be a noncommutative semilocal ring ($R/J(R)$ is a semisimple ring). Then there exists a unique determinant map from

$$\det : GL_n(R) \rightarrow R^\times / W(R)$$

where:

1. $\det(e) = 1$ for every elementary matrix e ,
2. if $\rho = \text{diag}(r, 1, \dots, 1)$ and $g \in GL_n(R)$, then $\det(\rho \cdot g) = r \cdot \det(g)$. Then show that $\det$ is a group homomorphism: $\det(gh) = \det(g)\det(h)$.

In particular, when reducing to $K_1(R)$:

$$K_1(R) \cong R^\times / W(R)$$

Proof :

Since R is a semilocal ring, every matrix in $GL_n(R)$ can be expressed as a product of elementary matrices. Define the map $\det : GL_n(R) \rightarrow R^\times / W(R)$ as follows. For any elementary matrix e , we set $\det(e) = 1$. For a diagonal matrix $\rho = \text{diag}(r_1, r_2, \dots, r_n)$, define

$$\det(\rho) = r_1 r_2 \cdots r_n \cdot W(R)$$

For a general matrix $g \in GL_n(R)$, express g as a product of elementary matrices and diagonal matrices, and define $\det(g)$ as the product of the determinants of these factors. As any matrix can be written as product of elementary matrices, this map is the unique determinant map (if it satisfies the determinant properties). And indeed, let $\rho = \text{diag}(r, 1, \dots, 1)$ and $g \in GL_n(R)$. Express g as a product $g = e_1 e_2 \cdots e_k \cdot d$, where each e_i is an elementary matrix and d the diagonal matrix. Then $\rho \cdot g = \rho \cdot e_1 e_2 \cdots e_k \cdot d$. Then this is equivalent to multiplying the first row of g by r , which multiplies the determinant by r . Hence,

$$\det(\rho \cdot g) = r \cdot \det(g)$$

Next, we must show $\det$ is a group homomorphism

$$\det(gh) = \det(g)\det(h) \text{ for all } g, h \in GL_n(R)$$

It suffices to establish this for elementary and diagonal matrices. For elementary matrices e_1 and e_2 , we have $\det(e_1 e_2) = 1 = \det(e_1)\det(e_2)$. For diagonal matrices $d_1 = \text{diag}(r_1, \dots, r_n)$ and $d_2 = \text{diag}(s_1, \dots, s_n)$, we have $\det(d_1 d_2) = \det(\text{diag}(r_1 s_1, \dots, r_n s_n)) = r_1 s_1 \cdots r_n s_n \cdot W(R) = (r_1 \cdots r_n \cdot W(R))(s_1 \cdots s_n \cdot W(R)) = \det(d_1)\det(d_2)$.

Finally, by definition $K_1(R)$ is the abelianization of $GL(R) = \varinjlim_n GL_n(R)$. The determinant map we constructed gives a homomorphism from $GL_n(R)$ to $R^\times/W(R)$ for each n , which are compatible under stabilization. Since $R^\times/W(R)$ is abelian and the determinant map captures exactly as we saw in example 3.1

$$K_1(R) \cong R^\times/W(R)$$

as we sought to show

Let us now focus on projective modules.

3.2 K_1 of C^* -algebra

here

3.2.1 Basics of C^* algebras

Definition 3.2.1: C^* -Algebra

A C^* -algebra A is an algebra over $\mathbb{C}$ with norm $a \mapsto \|a\|$ and involution $a \mapsto a^*$ such that A is complete with respect to the norm, $\|ab\| \leq \|a\|\|b\|$, and $\|a^*a\| = \|a\|^2$ for all $a, b \in A$

A C^* -algebra is called *unital* if it has a multiplicative identity. It is called *separable* if it contains a countable dense subset.

Given the axioms, it is easy to check that $\|a\| = \|a^*\|$.

Definition 3.2.2: $*$ -Homomorphism

Let A, B be C^* -algebras. Then a map $\varphi : A \rightarrow B$ is called a $*$ -homomorphism if it is a ring homomorphism that satisfies $\varphi(a^*) = \varphi(a)^*$. If A, B are unital and $\varphi(1_A) = 1_B$, then φ is called *unital* (or *unital preserving*)

A sub- C^* -algebra of A is a non-empty subset $B \subseteq A$ if it is closed under the 4 operations of A (addition, multiplication, adjoint, and scalar multiplication), as well as being a closed subset of A (so that it remains complete under the norm). If we simply have a sub $*$ -algebra of a C^* -algebra, then taking the norm closure gives us a sub C^* -algebra (since the 4 operations are continuous with respect to $\|\cdot\|$). Similar natural elaborations occur for generating C^* -algebra from a subset $S \subseteq A$. Quotients and products of C^* -algebras naturally follow the discussion from Banach Spaces (see Functional Analysis notes for details).

An important characterization of C^* -algebras is their “geometric” dual

Theorem 3.2.1: Gelfand-Naimark Duality

Let A be a C^* -algebra. Then there exists a Hilbert space H and an isometric $*$ -homomorphism $\varphi : A \rightarrow B(H)$. Thus every C^* -algebra is isomorphic to a sub C^* -algebra of $B(H)$. If A is separable, then H can be chosen to be a separable Hilbert Space

Proof :

see notes on functional Analysis, Gelfand Duality ([4]). The high-level idea is that we may view A as a vector-space equipped with a suitable inner product that comes from the positive linear functionals on A , which then we may complete. The rest is similar to Yoneda's Embedding theorem (recall the special case of a ring R being embedded in $B(R) = \text{End}_{\mathbf{Ab}}(R)$)

As mentioned, we shall consider C^* -algebras to not necessarily be unital. However, we may universally associate a unique unital C^* -algebra $\tilde{A}$ to a non-unital one A , with the characterization that $\tilde{A}/A \cong \mathbb{C}$.

Proposition 3.2.1: Unitization

Let A be a C^* -algebra. Then there exists a unital C^* -algebra $\tilde{A}$ such that $\tilde{A}/A \cong \mathbb{C}$. Furthermore, unitization is functorial.

Proof :

Let

$$\tilde{A} = A \oplus \mathbb{C}$$

with operation

$$(a, \alpha) \cdot (b, \beta) = (ab + \beta a + \alpha b, \alpha\beta) \quad (a, \alpha) = (a^*, \bar{\alpha})$$

and addition/scalar-multiplication is done coordinate wise. Then it is enough now to show that $(0, 1)$ is the unit in $\tilde{A}$. The rest of the properties now easily follow.

If A is already unital, then we get $\tilde{A} \cong A \oplus \mathbb{C}$ satisfies this property.

Every C^* -algebra naturally induces a new C^* -algebra $M_n(A)$ known as its *matrix algebra*. Each element of $M_n(A)$ is an $n \times n$ matrix whose elements are in A , addition, scalar multiplication, and involution is done term-wise, multiplication is matrix multiplication, and norm is the max of all the norms on the entries. With this norm, a function $f : X \rightarrow M_n(A)$ is continuous if and only if each component map $f_{ij} : X \rightarrow A$ is continuous. Naturally, if $\varphi : A \rightarrow B$ is a C^* -homomorphism, then $\varphi_n : M_n(A) \rightarrow M_n(B)$ given component wise operation is a C^* -homomorphism, and $M_n(-)$ is functorial.

Defining K group via Projections

First, we shall say two *elements* $a, b \in X$ are homotopic if there is a continuous path $[0, 1] \rightarrow X$ where $\gamma(0) = a$ and $\gamma(1) = b$ (this comes from the idea that X can be embedded in a natural hilbert space $L^2(X)$)

Definition 3.2.3: Projections And Equivalences

Let A be a C^* -algebra. The set of all projection of A shall be denoted $\mathcal{P}(A)$ (set of all elements such that $p = p^2 = p^*$). We shall define two equivalence relations on $\mathcal{P}(A)$:

1. (*von Neumann equivalence*) $p \sim q$ if there exists $v \in A$ such that $p = v^*v$ and $q = vv^*$
2. (*unitary equivalence*) $p \sim_u q$ if there exists a unitary element $u \in \mathcal{U}(\tilde{A})$ where $q = upu^*$.

Definition 3.2.4: Set of projections

Let

$$\mathcal{P}(A) = \mathcal{P}(M_n(A)) \quad \mathcal{P}_\infty(A) = \bigcup_n \mathcal{P}_n(A)$$

Then define $\sim_0$ on $\mathcal{P}(A)$ as follows

$$p \sim_0 q \iff v \in M_{m,n}(A), p = v^*v \text{ and } q = vv^*$$

now define $\oplus$ on $\mathcal{P}_\infty(A)$ via:

$$p \oplus q = \text{diag}(p, q) = \begin{pmatrix} p & 0 \\ 0 & q \end{pmatrix}$$

Then it is not hard to see this forms a semi-group structure

Proposition 3.2.2: Semigroup Of Projections

Let p, q, r, p', q' be projection in $\mathcal{P}_\infty(A)$ for some C^* -algebra A . Then:

1. $p \sim_0 p \oplus 0_n$ for every n
2. $p \sim_0 p'$ and $q \sim_0 q'$, then $p \oplus q \sim_0 p' \oplus q'$
3. $p \oplus q \sim_0 q \oplus p$
4. if p, q are projections in $\mathcal{P}_n(A)$ such that $pq = 0$, then $p+q$ is a projection and $p+q \sim_0 p \oplus q$
5. $(p \oplus q) \oplus r = p \oplus (q \oplus r)$

Proof :

all relatively simple, will leave for now

Definition 3.2.5: The Semigroup

Let $(\mathcal{P}(A), \sim_0, \oplus)$ be as in the above. Then

$$\mathcal{D}(A) = \mathcal{P}_\infty(A) / \sim_0$$

forms a semi-group where

$$[p]_{\mathcal{D}} + [q]_{\mathcal{D}} = [p \oplus q]_{\mathcal{D}}$$

Naturally, by the Grothendieck construction, this abelian group can be turned into a group. From this, we get the K_0 group.

3.2.2 K_1 theory of C^* algebras

here

Index

- Algebraic Line Bundle, 11
- Base Serre Cancellation, 8
- Bott Periodicity, 34
- Classifying Vector Bundles, 17
- Cofinal Monoid, 20
- Determinant Bundle, 14
- Elementary Matrices of $GL(R)$, 39
- First K-ring, 20
- Grothendieck Completion, 20
- H_0 Group, 25
- Hattori-Stallings Trace Map, 28
- Invariant Basis Property, 4
- K_0 of Rings, 21
- K_1 Group, 38
- K_1 groups, 39
- Mayer-Vietoris, 34
- Mayer-Vietoris Sequence, 33
- Milnor Patching, 9
- Morita Equivalence, 31, 32
- Picard Group, 12
- Picard Groups of Commutative Rings, 13
- Pierce's Theorem, 25
- Projective Module, 5
- Rank At Prime Ideal, 7
- Semi-local Ring, 41
- Serre-Swan Theorem, 11
- Stably Free, 4
- Trace and Character Representations, 29
- Vector Bundles on Sphere, 17

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